

Recent advances in design of magnetic functional alloys assisted by machine learning

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Pengqiang Hu,¹ Chao Zhou,^{1,a)} Sidan Ding,¹ Ruisheng Zhang,¹ Yizhe Ma,¹ Kun Wang,¹ Hongwen Chen,¹ Zhiyong Dai,¹ Rui Zhang,² Xiaoqin Ke,¹ Zhengming Zhang,³ Dunhui Wang,³ Yao Liu,⁴ Qiang Wang,⁵ Tie Liu,⁵ Wenli Pei,⁶ Tianli Zhang,⁷ Jingmin Wang,⁷ Jie Zhu,⁸ Xuexu Gao,⁸ Dan Huang,⁹ Yin Zhang,¹ Fanghua Tian,¹ Kaiyan Cao,¹ and Sen Yang^{1,a)}

AFFILIATIONS

- ¹School of Physics, MOE Key Laboratory for Nonequilibrium Synthesis and Modulation of Condensed Matter, Xi'an Jiaotong University, Xi'an 710049, China
- ²School of Physics and Centre for Research on Adaptive Nanostructures and Nanodevices, Trinity College, Dublin 2, Ireland
- ³Division of Microelectronic Materials and Devices, Hangzhou Dianzi University, Hangzhou 310018, China
- ⁴Frontier Institute of Science and Technology, State Key Laboratory for Mechanical Behavior of Materials, Xi'an Jiaotong University, Xi'an 710054, China
- ⁵Key Laboratory of Electromagnetic Processing of Materials (Ministry of Education), Northeastern University, Shenyang 110819, China
- ⁶Key Laboratory of Anisotropy and Texture of Materials (Ministry of Education), Northeastern University, Shenyang 110819, China
- ⁷School of Materials Science and Engineering, Beihang University, Beijing 100191, China
- ⁸State Key Laboratory for Advanced Metals and Materials, University of Science and Technology Beijing, Beijing 100083, China
- ⁹Shenyang National Laboratory (SYNL) for Materials Science, Institute of Metal Research, Chinese Academy of Sciences, Shenyang 110016, China

^{a)}Authors to whom any correspondence should be addressed: chao.zhou@xjtu.edu.cn and yang.sen@xjtu.edu.cn

ABSTRACT

Magnetic functional alloys (MFAs) constitute a vital category of smart materials, underpinning transformative technologies by the virtue of their emergent multifunctional properties. Nevertheless, the rational design of MFAs continues to pose significant challenges, stemming from expansive compositional space, complex interplay between magnetic behavior and crystal structure, and inherent limitations in conventional experimental and theoretical techniques. In the context of big data, machine learning (ML) has arisen as a revolutionary approach, facilitating the high-throughput discovery of new MFAs with tailored functionalities while simultaneously offering insights into their fundamental physical mechanisms. This review begins by outlining a general framework of ML-driven materials design, covering key algorithms and workflow strategies. Recent breakthroughs in the development of MFAs aided by ML are then highlighted, providing a comprehensive survey of data-driven advances in their design and application. Finally, we discuss persistent challenges and prospects arising from the integration of data-driven methodologies with physics-based models to propel this rapidly evolving field forward.

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TABLE OF CONTENTS

I. INTRODUCTION.....	2	1. ML model selection.....	4
II. GENERAL PARADIGM OF ML-AIDED MATERIALS DESIGN.....	2	2. ML model training.....	8
A. Database establishment.....	2	3. ML model evaluation.....	9
B. Feature engineering.....	4	D. Results validation.....	9
C. ML model construction.....	4	III. ML-AIDED DESIGN OF MAGNETIC FUNCTIONAL ALLOYS.....	10
		A. ML-aided design of magnetocaloric materials....	10

B. ML-aided design of magnetostrictive materials...	12
C. ML-aided design of magnetoresistance materials...	15
D. ML-aided design of shape memory alloys.....	17
IV. CHALLENGES AND OPPORTUNITIES.....	20
A. Dataset development.....	20
B. ML model interpretability.....	20
C. Multiscale complexity of MFAs.....	20
D. ML-aided inverse material design.....	20

I. INTRODUCTION

Materials constitute the foundation of modern technology, driving breakthroughs in energy systems, electronics, and biomedical applications. For decades, the pursuit of materials with optimized or emergent functionalities has relied heavily on iterative experimental synthesis and characterization. Such a process is often constrained by the inherent inefficiencies and costs to trial-and-error methodologies. While theoretical modeling and computational simulations have progressively shifted materials research toward a more predictive, knowledge-driven paradigm, persistent gaps remain in bridging atomic-scale insights with macroscale performance. For example, the first-principles calculations based on density functional theory (DFT) enable the prediction of electronic structures and material properties at the atomic level.¹ The Monte Carlo (MC) simulations reveal statistical behaviors aiding in understanding phase transitions,² while the molecular dynamics (MD) simulations provide insights into dynamic processes and thermal stability.³ Furthermore, the phase field (PF) model captures microstructural evolution,⁴ and the micromagnetic simulations optimize magnetic domain behaviors.⁵ These methods collectively accelerate the discovery and optimization of new materials.

In recent years, the maturity of artificial intelligence (AI) technology has further promoted the development of materials science. Especially, the combination of AI and big data have given birth to the “fourth paradigm” of science, i.e., big data-driven science.⁶ As a sub-field of AI, machine learning (ML) exhibits great vitality in materials science by combining the advantages of statistical techniques and computer systems.^{7,8} Meanwhile, a subset of ML that has demonstrated exceptional potential is deep learning (DL), which is increasingly utilized across diverse scientific and engineering fields because of its ability to model complex, high-dimensional data. Currently, a numerous of achievements on ML have been realized in materials science such as predicting mechanical performances,⁹ distinguishing phase components,¹⁰ designing photovoltaic materials,¹¹ etc. Correspondingly, we have compiled statistics on the publications of ML assisted materials design in the past 10 years. It can be also seen from left plot of Fig. 1(a) that the related work has shown explosive growth year by year.

Magnetic functional alloys (MFAs) have emerged as a transformative class of materials shown in Fig. 1(b), enabling critical technologies through their unique field-responsive behaviors, including magnetocaloric, magnetostrictive, magnetoresistive, and shape-memory effects. These multifunctional properties underpin their indispensable role in energy-efficient refrigeration, high-precision actuators, next-generation sensors, and advanced data storage systems.^{12,13} However, the advancement of MFAs encounters core obstacles, that is, the immense combinatorial design space and the

highly nonlinear structure–property relationships, dictated by intricate magnetic interactions and crystallographic symmetries, frequently surpass the capabilities of traditional experimental and computational methods. In contrast, ML offers inherent advantages for navigating this complexity, enabling high-throughput exploration of MFAs design rules beyond traditional paradigms. For instance, Lai *et al.*¹⁴ predicted a promising rare earth (RE)-free Fe₂P-type compound with a transition temperature of 97 K and a large magnetic entropy change (ΔS_M) of 7.5–11.5 Jkg^{−1}K^{−1} under a magnetic field change of 1 T, aided by artificial neural network, achieving the cryogenic magnetic refrigeration. Tian *et al.*¹⁵ proposed a noise-aware active learning strategy that can design the shape memory alloys possessing target latent heat through noisy data. Furthermore, Dai *et al.*¹⁶ developed a graph neural network model by combining the physical features of individual grains and their interactions, accomplishing an embedding of microstructure in polycrystalline materials. These findings demonstrate that ML technologies provide tremendous potential for rationally designing MFAs with desired properties. Utterly, we also provide a statistical analysis on the publications of ML-assisted MFAs design over the past decade. According to right plot of Fig. 1(a), we can find that the work on ML guided MFAs design is rather rare so far. Therefore, further effort is required to accelerate the design of MFAs based on ML techniques.

This review systematically examines recent advances in ML-aided design of MFAs. In Sec. II, we present a unified paradigm for ML-driven materials discovery, highlighting workflow design including database establishment, feature engineering, model construction and results validation. In Sec. III, we review the applications of ML across four classes of MFAs: (i) magnetocaloric materials, (ii) magnetostrictive compounds, (iii) magnetoresistance systems, and (iv) shape-memory alloys, with emphasis on structure–property relationship. Finally, the persistent challenges and emerging opportunities in data-driven MFAs development are critically assessed in Sec. IV, proposing pathways to integrate physics-aware ML with multiscale experimentation.

II. GENERAL PARADIGM OF ML-AIDED MATERIALS DESIGN

The general paradigm of ML-aided materials design is commonly categorized as four major modules: database establishment, feature engineering, ML model construction, and results validation, as exhibited in Fig. 2. To begin with, a database containing feature and target descriptors is prepared by excavating experimental or simulated data from relevant literature and public database. Then, the data pre-processing is performed, eventually reserving some key factors by feature engineering. Furthermore, the third step includes the construction of ML models based on the different algorithms, thereby selecting the optimal model by comparing predictive performance. Finally, the undeveloped composition with targeted properties can be predicted via the trained ML model. Next, a detailed introduction for four modules is provided.

A. Database establishment

Constructing high-quality database is a critical step in developing effective ML models. A well-designed database ensures that the models can be trained on accurate, representative and diverse data, which is essential for achieving robust performance.¹⁷ The process

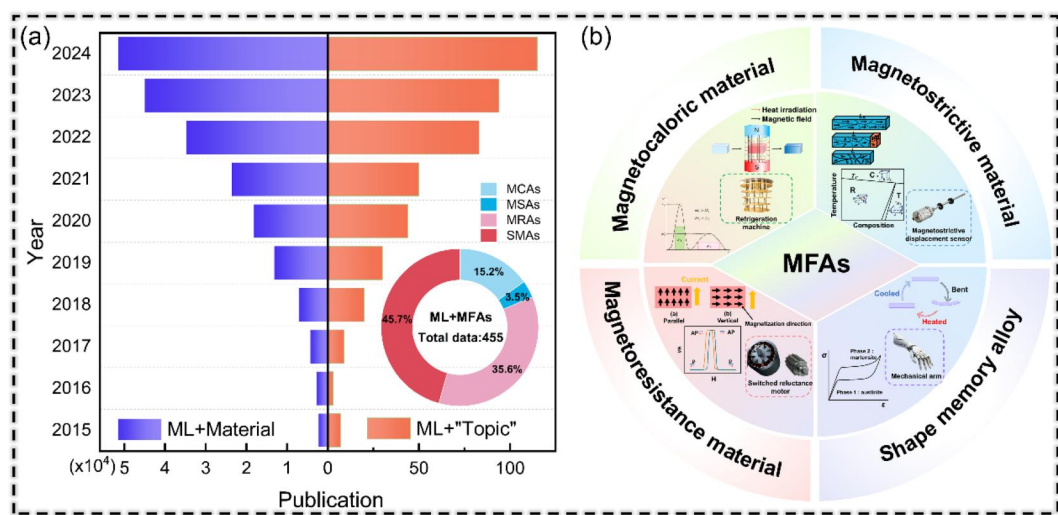


FIG. 1. (a) Annual publications retrieved with the keywords “machine learning + material” (left) and “machine learning + Topic” (right) from Web of Science over the past decade, where “Topic” refers to “magnetocaloric alloys (MCAs),” “magnetostrictive alloys (MSAs),” “magnetoresistance alloys (MRAs),” “shape memory alloys (SMAs).” The inset plot is the proportion of four types of functional materials design guided by ML. (b) Four classic MFAs including magnetocaloric material, magnetostrictive material, magnetoresistance material, and shape memory alloy, respectively.

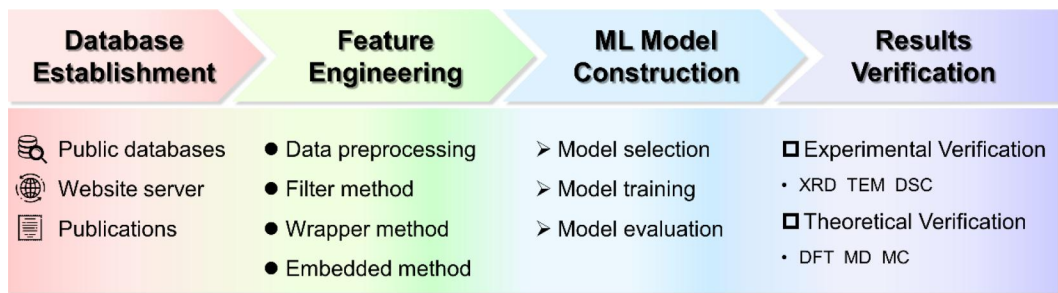


FIG. 2. The general paradigm of ML-aided materials design.

involves several key stages: data collection, preprocessing, and ongoing maintenance.

For materials science, the feature descriptors are mainly composed of extrinsic parameters (e.g., chemical composition, processing conditions, etc.) and intrinsic parameters (e.g., atomic parameters, structural parameters, etc.), whereas the predictive properties are used as the target descriptors such as ΔS_M , transition temperature, coercivity, remanence, etc. Currently, leveraging published literature is a prevalent strategy in the process of data collection for MAFs design.^{18–20} Moreover, some studies also built database by utilizing the data entries from experiments generated in their own laboratories such as the dataset of transformation temperatures, hysteresis and latent heat in NiTi-based alloys.^{15,21,22} Except gathering data from experiments, the high-throughput computational methods enable cost-effective assembly of extensive datasets like determining the thermodynamic descriptions by CALPHAD,²³ generating the eigenvalue of hysteresis through DFT,²⁴ and creating the features of mechanical properties based on phase field,²⁵ in shape memory alloys. Nevertheless, these approaches inevitably introduce hidden biases and data heterogeneity due to inconsistencies in data quality, experimental

conditions, measurement protocols, and simulation parameters. For these searching modes, it is worth mentioning that the text mining and the natural language processing (NLP) techniques are helpful for extracting the required information.²⁶ In addition, some authoritative databases are adopted like Materials Project,²⁷ AFLOWLIB,²⁸ Inorganic Crystal Structure Database (ICSD),²⁹ Open Quantum Materials Database (OQMD),³⁰ and AtomWork.³¹

Data preprocessing is crucial for cleaning and standardizing the data. This includes handling missing values, removing outliers, and normalizing features. For text data, preprocessing may involve tokenization, stemming, and removing stop words. In time-series data, techniques like resampling or interpolation are used to ensure consistency.³² The preprocessing also involves feature engineering, where the domain knowledge is applied to create meaningful features that can improve model performance. The ongoing maintenance is necessary to keep the database relevant and to date. This includes periodic updates to reflect changes in the problem domain, as well as continuous monitoring for data drift or degradation in model performance.

B. Feature engineering

A robust ML model strongly depends on a set of favorable feature descriptors. The lack/redundancy of features could give rise to underfitting/overfitting of ML model during training. To improve model performance and simplify data structure, a readable and meaningful representation of feature subset is necessary by converting the raw data. Feature engineering is an important technique in machine learning and data mining for removing impurities and redundancies from the raw data, involving data preprocessing and feature selection. Data preprocessing refers to transforming features and targets into an appropriate format that can be used for model training, such as the standardization and normalization for numerical features,³³ the conversion between visual images and corresponding pixel matrix,³⁴ and the transformation of categorical variable and vector based on one-hot encoding.³⁵ Additionally, selecting the most relevant feature combination is essential not only because it can enhance the predictive performance of ML models, but also because a well-prepared subset is crucial for shortening training time and costs. Feature selection is an effective method for avoiding curse of dimensionality, reducing over-fitting and making the model simple.

Based on the connection between ML algorithms and evaluation criteria, feature selection is generally categorized as filter, wrapper, and embedded methods.³⁶ The filter method can determine the correlation between features and target variables based on statistical techniques, like Pearson correlation coefficient (PCC), chi-square test, and mutual information. The wrapper method can detect the best subset by considering all possible feature combinations, such as recursive feature elimination and exhaustive screening. For the embedded method, the strategy is to evaluate feature importance based on the feature weight score during ML model training, such as decision tree (DT), random forest (RF), support vector machine (SVM), etc. Chen *et al.*³⁷ adopted a feature selection method combining feature importance based on RF and Bayesian optimization, retaining 135 features from the initial 378 features. Nevertheless, the advantages and disadvantages coexist in every feature selection. Hence, a combination of multiple feature selection strategies is more conducive to obtaining satisfactory results.

The hybrid method is expected to provide a good compromise between efficiency and effectiveness by exploiting different approaches. Wen *et al.*³⁸ employed a hybrid method involving correlation analysis and feature importance ranking. The highly correlated features exceeding 0.95 of correlation coefficient are filtered by PCC method, then ranking the relative importance of the remaining features based on a gradient boosting algorithm. Similarly, Liu *et al.*³⁹ employed a two-step screening approach to eliminate highly correlated feature pairs by calculating PCC and relative deviation of features. As exhibited in Fig. 3(a), Lu *et al.*⁴⁰ proposed an effectively approach through introducing a “last-place elimination” into the gradient boosting regression (GBR) model, which excluded the less-important features on targeted property, significantly improving the performance of the ML model. We recently adopted a process of feature selection assembled from three steps of filter, embedded, and wrapper methods, eventually determining 8 key factors from original feature pool with 34 features, as shown in Fig. 3(b).¹⁹

C. ML model construction

According to the prepared dataset and feature list via feature engineering, constructing the optimal ML model is required to precisely predict target attributes of materials. Through training samples and feature types, ML algorithms can be divided into supervised learning, unsupervised learning, semi-supervised learning and reinforcement learning. If a set of samples possessing the known input labels with the corresponding output labels is applied to the training of models, which is called supervised learning.⁴¹ On the contrary, unsupervised learning is the process of revealing hidden patterns in unlabeled datasets to fit models.⁴² Semi-supervised learning utilizes both labeled and unlabeled data to train models by combining supervised learning and unsupervised learning.⁴³ Another kind of reinforcement learning is an ML approach in which an agent learns to make decisions by performing actions and receiving rewards or penalties in a particular environment.⁴⁴ Among them, supervised learning is widely utilized in designing various material systems due to the advantages of handling regression and classification problems. In regression problems, the ML models are trained to predict continuous numerical values based on a set of input features. Classification problems refer to the procedure of determining the class of input samples by the trained data with the predefined labels based on the ML algorithms. Herein, the process of ML model construction is classified as three steps: ML model selection, ML model training, and ML model evaluation.

1. ML model selection

For the different tasks, every ML model exhibits dissimilar pros and cons. Hence, selecting an appropriate model is momentous based on task goal and sample dimension. Currently, a great deal of ML algorithms has been proffered to predict the target performances of materials. We next briefly introduce some representative ML models in the field of materials design, as depicted in Fig. 4. Meanwhile, Table I enumerates advantages and disadvantages of some common ML models in detail.

a. Kernel-based learning models. The main function of kernel-based methods is to map data from a low-dimensional space to a high-dimensional space, enabling linear methods to solve originally nonlinear problems in the high-dimensional space. Such learning method has been utilized in various ML models such as support vector machine (SVM),⁴⁵ Gaussian process regression (GPR),⁴⁶ and kernel ridge regression (KRR).⁴⁷

As one of the most powerful and versatile supervised learning models, the SVM is primarily used to tackle regression and classification tasks in materials science. The SVM can effectively discriminate between data points by constructing a hyperplane within a higher-dimensional feature space, further offering robust performance even with limited training data and handling complex, nonlinear relationships through the application of kernel tricks.⁴⁸ For the core principle of SVM regression and classification, the SVM regression aims to find a function that minimizes the distance between data points and hyperplane, while the SVM classification seeks to identify the maximum-margin hyperplane that separates different classes of data points, thereby maximizing the boundary between classes. It should

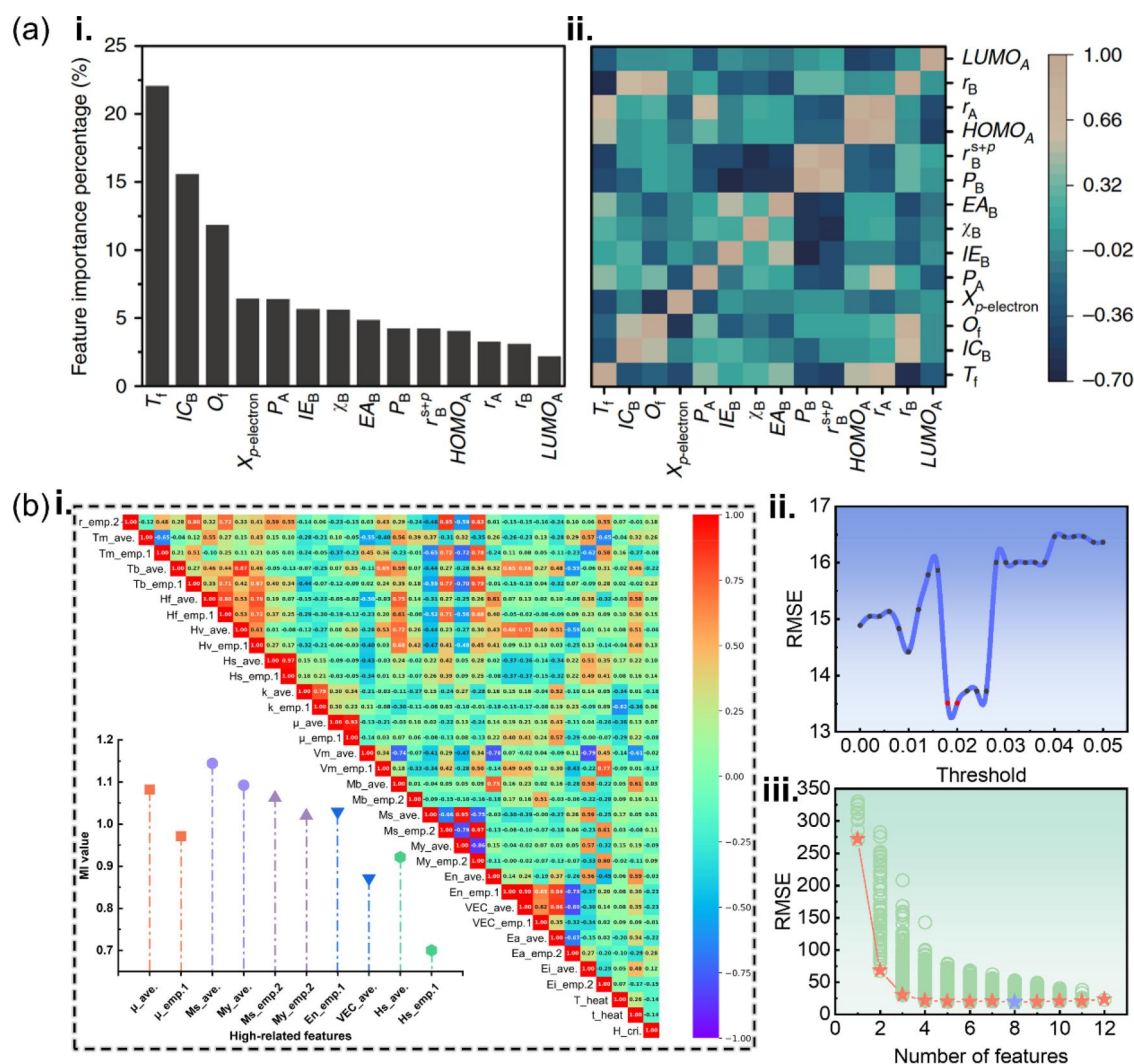


FIG. 3. The feature selection procedure. (a) (i) Feature importance ranking of 14 selected features using GBR algorithm. (ii) Heatmap of PCC matrix for the selected features. Reproduced with permission from Lu *et al.*, Nat. Commun. 9, 3405 (2018). Copyright 2018 Authors, licensed under a Creative Commons Attribution (CC BY) license.⁴⁰ (b) (i) PCC heatmap of the 34 features in the original feature pool. The lower-left plot is the mutual information value of high-correlated features. (ii) Feature selection by random forest, red solid points stand for the reserved subset under the best threshold. (iii) Exhaustive screening, blue solid star represents the best subset after 10-fold cross validation. Reproduced with permission from Hu *et al.*, Mater. Des. 252, 113799 (2025). Copyright 2025 Authors, licensed under a Creative Commons Attribution (CC BY) license.¹⁹

be mentioned that the kernel trick of SVM is vital for predicting material properties. The popular SVM kernels includes the linear, polynomial and Gaussian kernels (also called radial basis function (RBF) kernel).⁴⁹ Typically, a SVR-based model was applied to estimating the relative cooling power of doped manganites.⁵⁰ In a limited training data, SVR model can exhibit the better performance than ML models.

The GPR is a nonlinear and nonparametric regression method based on Bayesian probability theory, which assumes that the observed data points are drawn from a multivariate Gaussian distribution, defined by a mean function and a covariance (kernel) function.⁵¹ The GPR excels in capturing uncertainty and providing probabilistic

predictions, making it ideal for small datasets with noisy or sparse data. Owing to a huge popularity in materials design, GPR has been widely applied to predict material performances and optimize compositions. For instance, Dangwal *et al.*⁵² adopted the GPR model to predict the enthalpy of hydride formation in high entropy alloys for room-temperature hydrogen storage, achieving high accuracy compared to experimental and DFT values. Additionally, Thoppil *et al.*⁵³ utilized the GPR with combination kernels to predict mechanical properties of complex concentrated alloys, enabling the tracing of property evolution at each processing stage and facilitating high-throughput screening of alloy compositions with desired mechanical characteristics.

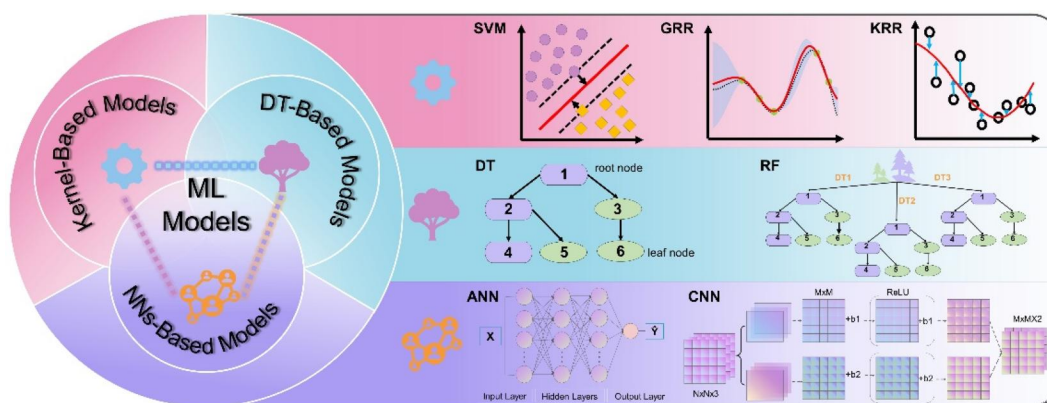


FIG. 4. Schematic of the common ML models in designing MFAs. Kernel-based learning models: support vector machine (SVM), Gaussian process regression (GPR), and kernel ridge regression (KRR). DT-based learning models: decision tree (DT), random Forest (RF). NNs-based learning models: artificial neural networks (ANN) and convolutional neural network (CNN).

The KRR belongs to a nonlinear regression technique combining ridge regression (L2 regularization) with the kernel trick, enabling robust modeling of complex relationships in high-dimensional datasets. By projecting input features into a reproducing kernel Hilbert space, the KRR captures nonlinear patterns while mitigating overfitting, making it particularly effective for materials datasets with limited samples but high complexity. The KRR has emerged as a powerful tool for discovering new materials. For example, Li *et al.*⁵⁴ employed the KRR and the extra trees classifier to predict the thermodynamic stability of perovskite oxides, achieving high accuracy with minimal computational cost compared to traditional DFT calculations. Similarly, Ahmad *et al.*⁵⁵ used the KRR to screen over 12 000 inorganic solid electrolytes for suppressing dendrite formation in lithium metal anodes. By predicting mechanical properties, the 20 promising anisotropic interfaces were identified, which could enable stable electrodeposition and high ionic conductivity.

The advantages of kernel-based models lie in their robustness, particularly when data are limited. They can provide accurate predictions for molecular and material properties. Moreover, kernel regression has CPU cost advantages compared to neural networks. However, a major disadvantage is that in high-dimensional feature spaces, Matern kernel-based methods may degenerate into low-order polynomial regressions, leading to a loss of expressive power.⁵⁶ Additionally, kernel-based models may lack insight into the importance of specific variables compared to other machine learning methods.

b. DT-based learning models. The decision tree (DT) model has emerged as robust learning in the machine learning toolkit, offering an intuitive flowchart-like structure for solving regression and classification problems. The DT is structurally consisted of internal nodes and leaves like a flowchart-like structure, in which each internal node in tree represents a logical test on a feature, each branch corresponds to an outcome of the test, and each leaf node holds a constant number (for regression trees) or a class label (for classification trees).⁵⁷ Given the strong interpretability and simple structure of DT,⁵⁸ research workers have paid considerable attention toward the field of DT-assisted material design.^{59,60} However, the DT is confronted with the

major issue that the model incline to overfitting easily with training.^{49,59} It is feasible to create a more powerful learning model by combining multiple weak learning models based on the concept of ensemble methods.^{61,62} Thus, a series of DT-based ensemble learning models are subsequently developed. In this section, we review some common DT-based ensemble learning models involving random forest (RF),⁶³ gradient boosting decision tree (GBDT),⁶⁴ and extreme gradient boost (XGBoost).⁶⁵

RF is an ensemble learning method, which can improve the accuracy and generalization ability of model by constructing multiple decision trees.⁶⁶ The core of RF is to produce decision trees in parallel on a bootstrap sample dataset via using bagging in RF, further integrating the ultimate predictions through voting (for classification) or averaging (for regression). Owing to feature importance ranking, RF algorithm is extensively used in designing various materials through aiding in feature selection.¹⁴ For the prediction of rhombohedral structures in M_2X_3 -type thermoelectrics, the RF model enabled feature selection in high-dimensional and small datasets, improving model interpretability and predictive accuracy.³⁷ Differing from bagging concept employed in RF, the GBDT belongs to a boosting model in ensemble learning. The boosting models are serially established by training multiple weak models via optimizing errors of every previous model.⁶⁷ The basic rule of GBDT is to train weak models by iteratively minimizing the negative gradients of loss function by gradient descent method at each round training, ultimately aggregating predictions from all weak models with certain weights to form the final prediction. Due to high prediction accuracy and good robustness, GBDT is utilized to process regression, classification, and ranking tasks in a variety of areas.^{68–70} Wei *et al.*⁷¹ achieved high accuracy of $R^2 = 0.9491$ in predicting electrochemical impedance spectroscopy for corrosion of high-entropy alloys by GBDT, in which the iterative boosting effectively captured complex relationships and reduces residuals, handling complex input–output relationships. XGBoost is essentially an efficient implementation of GBDT, with higher scalability in speed, generalization ability, and accuracy.⁷² Furthermore, XGBoost also shows some other strengths such as automatically processing the missing values by the sparse-aware

TABLE I. Common machine learning models in magnetic functional alloys design.

Algorithm	Description	Advantage	Disadvantage
Support vector machine (SVM)	Minimize data point-hyperplane distance in regression, determine maximum-margin separating hyperplane in classification.	Robust with limited training data, handle complex nonlinear relationships via kernel tricks.	Unfit for very large datasets (slow training), performance drops with noisy/overlapping data, poor interpretability.
Gaussian process regression (GPR)	Non-linear nonparametric Bayesian regression, assume observations follow multivariate Gaussian distribution.	Provide probabilistic predictions with uncertainty tracking, ideal for small noisy/sparse datasets.	High computational cost for large datasets, performance highly sensitive to kernel function selection, struggle in high-dimensional input spaces.
Kernel ridge regression (KRR)	Non-linear regression combining L2-regularized ridge regression and kernel trick, map features to reproducing kernel Hilbert space to capture nonlinear patterns.	Enable robust complex relationship modeling on high-dimensional data, mitigate overfitting for small complex materials datasets.	High computational cost for very large dataset, requires careful kernel and regularization parameter tuning.
Random forest (RF)	Ensemble learning method built on multiple decision trees, integrate bagging and random feature selection.	Perform well on high-dimensional data and nonlinear relationships, robust to noise and outliers via averaging/voting mechanism.	Less interpretable than single decision tree, requires more memory and computing resources for large forests.
Gradient boosting decision tree (GBDT)	Sequential ensemble learning methods that build new trees to correct previous residual errors of trees through gradient descent.	Deliver higher accuracy than RF on structured/tabular data, address nonlinear relationships and feature interactions.	Prone to overfitting with excessive tree count or tree depth, require careful hyperparameter tuning.
Extreme gradient boost (XGBoost)	Optimized scalable GBDT implementation, add L1/L2 regularization for reduced overfitting and better generalization.	Achieve state-of-the-art performance on structured/tabular data, natively handle missing values via learned default split directions.	Require more hyperparameters, easily overfit on very small or noisy datasets.
Artificial neural network (ANN)	Brain-inspired computational model consisting of input layer, multiple hidden layers and output layer, each neuron carries a nonlinear activation function.	Approximate any complex nonlinear function, process high-dimensional unstructured data like images, text and audio.	Demand massive labeled training data for good generalization, computationally expensive and slow to train without GPUs.
Convolutional neural network (CNN)	Specialized ANN designed for grid-like data (e.g., images), use learnable-filter convolutional layers for spatial feature extraction and pooling layers for dimensionality reduction.	Deliver robust performance in image classification, object detection and segmentation, drastically cuts parameter count.	Require large labeled datasets and GPU resources for training, limited effectiveness on nonimage data.
Graph neural network (GNN)	Neural architecture for graph-structured data, update node representations via neighboring information aggregation, learn node-level and graph-level embeddings for downstream tasks.	Natively process irregular non-Euclidean data such as molecules, crystals and social networks, capture relational dependencies and topological structures.	Suffer from over-smoothing (indistinguishable node features) with excessive stacked layers, high computational cost for large graphs due to neighborhood explosion.

algorithm and reducing the model overfitting through adding the regularization term to the objective function. In TiNiCu elastocaloric materials, a minimal thermal hysteresis of 1.2 K was predicted by the XGBoost model via handling of small, irregular datasets, and integration of physical features.²⁴

c. NNs-based learning models. Neural networks (NNs) are forceful machine learning tools inspired by biological neurons, capable of modeling complex, nonlinear relationships in high-dimensional data. In materials science, NNs have been widely applied to optimize properties involving phase stability, mechanical strength, and electronic

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behavior. We introduce some applications of classical NNs in materials design, including the artificial neural network (ANN), the convolutional neural network (CNN) and the graph neural network (GNN).

The ANN is a computational model constructed by referencing the way biological neural networks process information in the human brain.⁷³ Such algorithm is composed of three primary layers, i.e., the input layer, the hidden layer and the output layer, which are interconnected by the weights between nodes (neurons).⁷⁴ The input layer receives raw data, where each node represents a feature of the dataset. The hidden layers, which can be multiple, transform the input data through weighted connections and activation functions (e.g., ReLU, sigmoid) to capture complex patterns. Eventually, the output layer produces the model predictions, such as class labels or continuous values, depending on the task. For addressing complex linear and nonlinear relationships, ANN is provided with large superiority due to its learning ability and generalization capability. Based on this model, Radhamani *et al.*⁷⁵ clarified the complex interaction between electronic factors and composition in governing the phase transformation temperatures in NiTiCu alloys.

The CNN is a subset of neural networks that is assembled by convolutional, pooling, and fully connected layers. The convolutional layer is the core component in CNN, which can extract spatial features using learnable filters, further detecting patterns like edges or textures while preserving spatial relationships. The pooling layer reduces feature map dimensions through operations such as max or average pooling, enhancing computational efficiency and reducing overfitting. For the fully connected layer, its function is to integrate the extracted features to produce outputs by connecting every neuron to all activations. In design of shape memory alloy-based wire actuator, 1-dimensional CNN was employed to determine position estimation by using self-sensing properties, which showed faster inference with comparable accuracy, demonstrating that it is suitable for real-time applications.⁷⁶ In addition, considering that these layers enable CNN to hierarchically learn and process complex patterns, CNN has been extensively used in image recognition,⁷⁷ speech recognition,⁷⁸ and natural language processing.⁷⁹

The GNN is a vigorous class of DL architectures specifically designed to analyze and learn from graph-structured data, where the entities are represented as nodes and relationships as edges.⁸⁰ Recently, GNN has gained significant attention due to their ability to model complex relational data. For example, Kipf and Welling⁸¹ introduced the graph convolutional network (GCN), which generalizes convolutional operations to graphs, achieving the state-of-the-art performance in semi-supervised node classification. Similarly, Veličković *et al.*⁸² proposed the graph attention network (GAT) that use attention mechanisms to weigh the importance of neighboring nodes, improving model interpretability and performance. In addition, Stokes *et al.*⁸³ used GNN to discover a novel antibiotic by predicting molecular properties. Nevertheless, the GNN also faces some challenges such as over-smoothing (node features becoming indistinguishable) and over-squashing (information bottlenecks in dense graphs).

It should be mentioned that a graph-based deep learning method, crystal graph convolutional neural networks (CGCNN), is widely adopted to predict material properties directly from graph-like representations of crystal structures.^{84–86} Recently, a large number of studies have been focused on developing CGCNN models to discover

novel magnetic materials, predict magnetic moments, Curie temperatures, and anisotropy. Representatively, Xia *et al.*⁸⁷ presented an ML-guided feedback framework integrating CGCNN, adaptive genetic algorithm and DFT to successfully identify and synthesis Fe₃CoB₂, a rare-earth-free magnet with high anisotropy and saturation polarization. The generative GNNs and high-throughput DFT were created by Elrashidy to discover 2D magnets.⁸⁸ They confirmed 167 stable magnetic monolayers with a magnetization $\geq 0.5\mu_B$ and energy above the convex hull < 0.3 eV/atom from 11 100 AI-generated candidates. Based on the CGCNN, materials graph network and random forests using DFT data from the Materials Project database, Kaba *et al.*⁸⁹ predicted large-magnetic-moment materials. The models were applied to the ICSD database, identifying 15 promising candidates theoretically. They concluded that magnetic moment prediction proved more challenging than other properties like formation energy, partly due to ferromagnetic bias in training data. These findings demonstrate that CGCNN and related GNNs are becoming state-of-the-art patterns for high-throughput screening in magnetic materials discovery.

2. ML model training

ML models training is a fundamental process that involves optimizing model parameters to minimize a predefined loss function, typically using gradient-based optimization techniques such as stochastic gradient descent (SGD) or its variants (e.g., Adam and RMSprop). This process requires careful partitioning of data into training, validation, and test sets to ensure that the model generalizes well to unknown data. However, the performance of ML models is highly dependent on hyperparameters, which are configuration settings that govern the training process and model architecture. The hyperparameters include learning rate, batch size, number of layers, and regularization parameters. Hyperparameter optimization (HPO) is therefore a critical step in building effective ML models.

Traditional methods for HPO include grid search and random search. Grid search exhaustively evaluates all possible combinations of hyperparameters within a predefined range, while random search samples hyperparameters randomly from the search space. Although grid search is thorough, it is computationally expensive, especially for high-dimensional hyperparameter spaces. Random search, on the other hand, has been shown to be more efficient in practice, as it often finds good hyperparameters with fewer evaluations.⁹⁰ However, both methods can be inefficient for large-scale models and datasets.

To address these limitations, more advanced HPO techniques have been developed. Bayesian optimization is a popular approach that builds a probabilistic model of the objective function (e.g., validation accuracy). By iteratively updating the probabilistic model, Bayesian optimization balances exploration and exploitation, making it highly efficient for expensive-to-evaluate functions.⁹¹ Another approach is evolutionary algorithms, which mimic natural selection to evolve a population of hyperparameter configurations over multiple generations. These methods are particularly useful for optimizing complex, nondifferentiable objectives.

In recent years, the neural architecture search (NAS) has emerged as a powerful technique for automating the design of NNs architecture. The NAS methods, such as reinforcement learning-based approaches⁹² and differentiable NAS,⁹³ have achieved state-of-the-art results in tasks like image classification and natural language processing. These methods not only optimize hyperparameters but

also discover novel architectures that outperform manually designed ones. However, the NAS can be computationally intensive, requiring thousands of GPU hours, which has led to the development of more efficient techniques like weight-sharing and progressive search.

Another important advancement in HPO is multi-fidelity optimization, which reduces computational costs by evaluating hyperparameters on lower-fidelity approximations of the objective function. For instance, the Hyperband dynamically allocates resources to promising configurations and early terminates poorly performing ones, significantly speeding up the optimization process.⁹⁴ Similarly, the successive halving and BOHB (Bayesian Optimization and Hyperband) combine the strengths of Bayesian optimization and multi-fidelity methods to achieve efficient and robust HPO.

During the process of HPO, one major issue is the risk of overfitting to the validation set, especially when the search space is large or the dataset is small. Cross-validation and robust evaluation metrics are essential to mitigate this risk. Additionally, the computational cost of HPO remains a significant barrier for large-scale models and datasets. Hence, transfer learning and meta-learning have been proposed as strategies to leverage knowledge from previous tasks to accelerate HPO. For example, the meta-learning techniques use historical data from similar tasks to warm-start the optimization process, reducing the required number of evaluations.⁹⁵

3. ML model evaluation

The evaluation of ML models is essential to ensure their performance, generalizability, and reliability. The process involves assessing models on unknown data using task-specific metrics. For the classification problems, the common metrics include accuracy, precision, recall, F1-score, and ROC-AUC. The accuracy measures the correct predictions, yet which can be misleading in imbalanced datasets. The precision and the recall focus on the positive class identification, while the ROC-AUC evaluates the class separation across thresholds. Accordingly, the metrics of the regression problems have the mean squared error (MSE), the root mean squared error (RMSE) the mean absolute error (MAE), and the R-square (R^2). Some of evaluated criteria for classification and regression problems are listed in Table II.

The robust evaluation requires the proper data splitting into training, validation, and test sets. Particularly, the k-fold cross-validation is widely used to reduce the variability in performance estimates. The k-fold cross-validation refers that the data are partitioned into k subsets, subsequently training models on k-1 subsets, and validating on the remaining subset, repeating k times.⁹⁶

The model evaluation can also identify issues like overfitting and underfitting. Overfitting occurs when the models perform well on the training set but poorly on test set. Regularization and early stopping are good techniques for mitigating the overfitting. Conversely, the underfitting is caused by overly simplistic models, which can be addressed by increasing complexity or feature engineering. Moreover, interpretability and fairness are increasingly important. Currently, some techniques like SHapley Additive exPlanations (SHAP) and Local Interpretable Model-Agnostic Explanations (LIME) are applied to explain model predictions, enhancing transparency.⁹⁷ The fairness metrics is used to ensure equitable outcomes, such as demographic parity, and detect biases.⁹⁸

TABLE II. Evaluated criteria for classification and regression problems.

Type	Evaluated criteria	Formula
Classification	Accuracy	$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$
	Precision	$Precision = \frac{TP}{TP + FP}$
	Recall	$Recall = \frac{TP}{TP + FN}$
	F1-score	$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$
	ROC-AUC	$Sensitivity = \frac{TP}{TP + FN}$
		$Specificity = \frac{TN}{FP + TN}$
Regression	Mean squared error (MSE)	$MSE = \frac{1}{m} \sum (y_i - y_{pre})^2$
	Root mean squared error (RMSE)	$RMSE = \sqrt{\frac{1}{m} \sum (y_i - y_{pre})^2}$
	Mean absolute error (MAE)	$MAE = \frac{1}{m} \sum y_i - y_{pre} $
	R-square (R^2)	$R^2 = \frac{\sum (y_{pre} - \bar{y})^2}{\sum (y_i - \bar{y})^2}$

D. Results validation

The successful integration of ML in materials design hinges on rigorous validation through analyzing the predictive results. Generally, the predictive results are validated based on experimental measurements/theoretical simulations. Such process can be summarized into the following steps. First, a virtual composition space corresponding to tasks is obligatory to realize high throughput screening. A series of target properties of materials are subsequently predicted by the ML models based on the designed virtual composition space. Eventually, the compositions picked from predicted samples are arranged for experimental preparation and characterization/theoretical simulations. Currently, advanced robotic platforms have enabled autonomous experimental validation, where ML models iteratively guide synthesis and testing cycles to refine predictions.⁹⁹ For theoretical validation, atomic-scale insights can be provided to complement experiments by various techniques like DFT calculations, and MD, MC, PF simulations, etc. These methods are particularly valuable when experimental data are scarce or when probing fundamental mechanisms underlying ML predictions.¹⁰⁰ Recent studies demonstrate how hybrid approaches combining ML with DFT can accurately predict novel materials while reducing computational costs.¹⁰¹

A critical challenge remains in bridging the gap between predicted and observed material properties. Discrepancies often arise from limitations in training datasets or incomplete physical descriptors in ML models. To address this issue, modern frameworks incorporate uncertainty quantification and active learning to prioritize high-value experiments. In addition, the merging techniques also integrate explainable AI to interpret ML predictions and identify physically meaningful patterns.

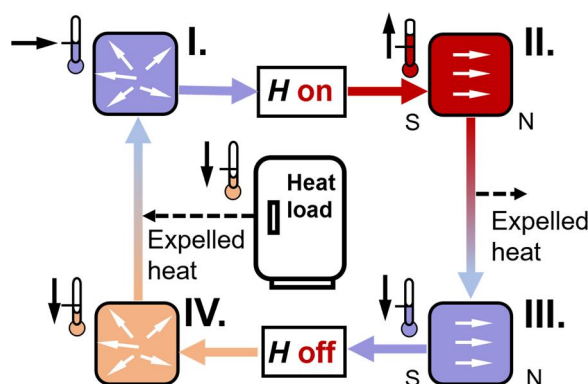


FIG. 5. Schematic for mechanism of magnetocaloric effect.

III. ML-AIDED DESIGN OF MAGNETIC FUNCTIONAL ALLOYS

The ML has become increasingly popular in materials research, achieving significant progress in magnetic functional materials design. Especially, the ML models show the strong power in optimizing material properties, predicting performance and accelerating the discovery of novel compositions.¹⁰² In this section, we provide an overview of some classic magnetic functional materials and the application of

machine learning in this field, including magnetocaloric, magnetostrictive, magnetoresistance materials, and shape memory alloys.

A. ML-aided design of magnetocaloric materials

Magnetocaloric effect (MCE), which is observed by Weiss in Ni,¹⁰³ is a thermal phenomenon that magnetic materials undergo the release and absorption of heat due to the change of magnetic ordering upon applying or removing magnetic field, as depicted in Fig. 5. Based on this effect, the concept of magnetic refrigeration was proposed by Debye¹⁰⁴ and Giauque,¹⁰⁵ subsequently achieving cooling purpose by using paramagnetic salts experimentally.¹⁰⁶ In the past decades, a large number of refrigerants are developed like Gd-Si-Ge-based,¹⁰⁷ Mn-Fe-P-As-based,¹⁰⁸ La-Fe-Si-based,¹⁰⁹ and Ni-Mn-based compounds.¹¹⁰ Meanwhile, the development of magnetic refrigeration machines has also shown explosive growth due to higher energy efficiency, environmental friendliness and reduced noise.¹¹¹

In recent years, the design of magnetocaloric materials for efficient cooling applications has been significantly advanced by ML techniques. It is reported that researchers have employed different ML models to predict ΔS_M and MCE in many classical magnetocaloric materials. For example, Zhao *et al.*¹¹² determined a high-efficiency corrosion inhibitor for La(Fe, Si)₁₃ magnetic refrigeration materials using linear regression due to the advantages of model with much simpler and easier, as well as harder overfitting. Figure 6(a) exhibits measured-predicted pH value, corrosion current density J

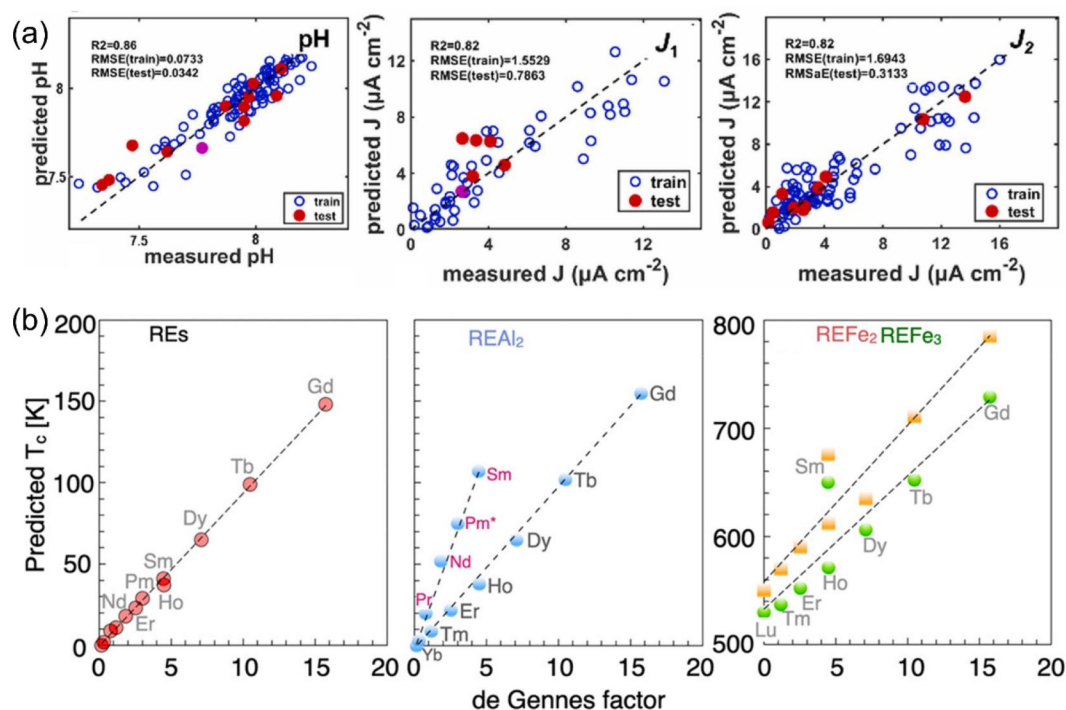


FIG. 6. (a) Measured-predicted pH value, corrosion current density J_1 value ($\text{Na}_2\text{HPO}_4 \cdot 12\text{H}_2\text{O} < 3.5 \text{ g} \cdot \text{L}^{-1}$), and J_2 value ($\text{Na}_2\text{HPO}_4 \cdot 12\text{H}_2\text{O} \geq 3.5 \text{ g} \cdot \text{L}^{-1}$) by linear regression model. Reproduced with permission from Zhao *et al.*, Corros. Sci. **216**, 111115 (2023). Copyright 2023 Elsevier.¹¹² (b) Linear relation between the predicted T_c and the de Gennes factor $[(g-1)^2 J(J+1)]$ for RE-metals, rare-earth dialuminides, REAl_2 , and rare-earth-iron, REFe_2 and REFe_3 compounds, where g and $J(L \pm S)$ are the Landé g -factor and total angular momentum quantum number, respectively. Reproduced with permission from Singh *et al.*, Chem. Mater. **35** 6304–6312 (2023). Copyright 2023 American Chemical Society.¹¹³

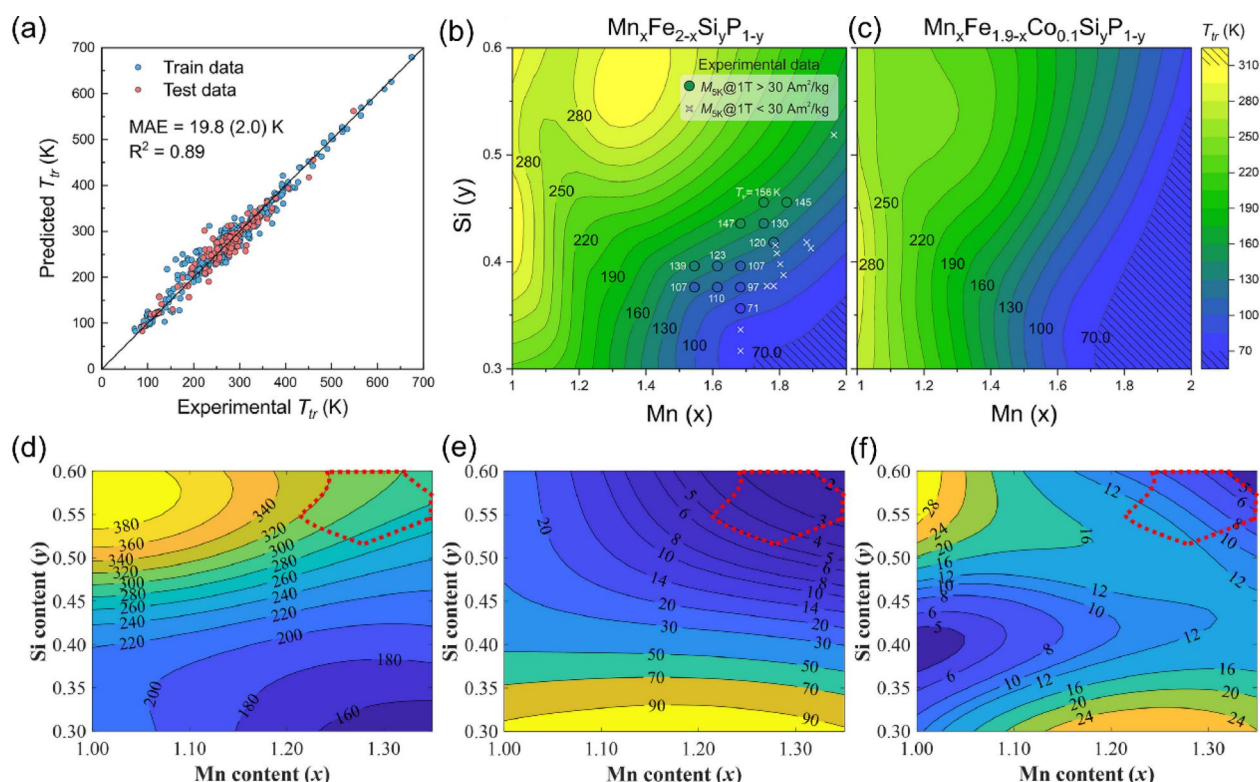


FIG. 7. (a) Comparison of the experimental and predicted transition temperatures using ANN for Fe₂P-based compounds. Colormaps of the predicted transition temperatures for (b) Mn_xFe_{2-x}Si_yP_{1-y} and (c) Mn_xFe_{1.9-x}Co_{0.1}Si_yP_{1-y} systems. Reproduced with permission from Lai *et al.*, *Acta Mater.* **232**, 117942 (2022). Copyright 2022 Elsevier.¹⁴ Predicted effect of Mn and Si on (d) T_C , (e) thermal hysteresis, and (f) ΔS_M for Mn_xFe_{2-x}P_{1-y}Si_y alloys. Reproduced with permission from Tu *et al.*, *J. Mater. Sci. Technol.* **96**, 241–247 (2022). Copyright 2022 Elsevier.¹⁶

value (J_1 , J_2) by linear regression model. The model showed a good performance with R^2 of 0.86 for Ph value. However, the performance of model in predicting corrosion current density (J) was suboptimal due to the high sensitivity of J value to concentration changes. They thus segmented the J dataset into two subsets: J_1 ($\text{Na}_2\text{HPO}_4 \cdot 12\text{H}_2\text{O} < 3.5 \text{ g}\cdot\text{L}^{-1}$) and J_2 ($\text{Na}_2\text{HPO}_4 \cdot 12\text{H}_2\text{O} \geq 3.5 \text{ g}\cdot\text{L}^{-1}$), significantly enhancing modeling accuracy with R^2 of 0.82. The designed corrosion inhibitor, composed of Na_2MoO_4 and Na_2HPO_4 , effectively protected La(Fe, Si)₁₃, maintaining a smooth surface and large, stable MCE.

Tree-based models offer high interpretability and computational efficiency for predicting material properties, effectively capturing nonlinear relationships even with limited datasets. Singh *et al.*¹¹³ presented a physics-informed RF model to unveil patterns in the chemical-to-Curie temperature (T_C) relationship of RE-based metals. Additionally, they examined the correlation between T_C and the de Gennes scaling factor in rare-earth intermetallics, analyzing its breakdown below a critical rare-earth content. The results illustrated that a strong linear correlation between the de Gennes factor and the predicted T_C was confirmed in REAl₂, REFe₂, and REFe₃ compounds [Fig. 6(b)]. In contrast, it is demonstrated that this scaling breaks down in RE-poor, transition-metal-rich systems like RECo₅ and RE₂X₁₇, where T_C no longer follows the de Gennes factor. Zhang *et al.*¹¹⁴ predicted the

maximum ΔS_M and T_C of La(Fe, Si/Al)₁₃-based materials based on chemical composition by GBRT algorithm, aiding efficient composition design for magnetic refrigeration applications. In order to enhance model performance, incorporating more physically meaningful features like lattice constants and saturation magnetization may be a reliable way.

NNs-based models excel at automatically learning complex, high-dimensional structure–property relationships from materials data, enabling accurate and efficient property prediction without relying on handcrafted features. Maiorino *et al.*¹¹⁵ assessed the thermodynamic performance of La–Fe–Co–Si alloy based on a multi-layer perceptron (MLP) with two inputs, one hidden layer, and two outputs, which further accurately predicted both the isothermal entropy change and adiabatic temperature change. Furthermore, Lai *et al.*¹⁴ identified a promising RE-free composition of Mn_{1.7}Fe_{0.3}P_{0.63}Si_{0.37} guided by ANN model, exhibiting a tunable transition temperature from 97 to 73 K under 1 T via minor Co substitution, and delivering a large ΔS_M of 7.5–11.5 J kg^{−1} K^{−1} below 100 K. Figure 7(a) presents the predictive performance of the ANN trained on the final Fe₂P-based dataset with a MAE of 19.8 (2.0) K and a R^2 of 0.89. A colormap of the predicted transition temperatures are exhibited in Fig. 7(b), where the annealing temperature and time are fixed at 1373 K and 5 h, respectively. Experimental samples are marked as colored dots, and

those with low magnetization ($M_{5K@1T} < 30 \text{ Am}^2\text{-kg}^{-1}$) are labeled with crosses to indicate the limitation on reduction of transition temperature. As shown in Fig. 7(c), a small Co addition expands the low-transition temperature region for compounds. Similarly, Tu *et al.*¹¹⁶ designed Mn-Fe-P-Si compound with large ΔS_M , low hysteresis through NNs model, simultaneously offering a compositional window with outstanding MCE. As displayed in Figs. 7(d) and 7(e), the composition-dependent trends for T_C , hysteresis and ΔS_M provide a basis for designing a series of Mn-Fe-P-Si alloys. By systematically varying composition within the identified windows, a gradual modulation of T_C can be achieved while maintaining a large ΔS_M , thereby expanding the operational temperature span.

Considering the remarkable MCE in Heusler alloys, some of these intermetallics have become promising candidates for magnetic refrigeration. Gharaibeh *et al.*¹¹⁷ trained an unsupervised learning model to predict Heusler alloys with giant MCE. They identified several candidates such as MnCoGe-based, $\text{Ni}_{43}\text{Mn}_{46}\text{Sn}_{11}$ and $\text{Ni}_{50}\text{Mn}_{34}\text{In}_{16}$ by considering factors of magnetic entropy changes and magnetostructural transitions. In addition, Baigutlin *et al.*¹¹⁸ predicted structural and magnetic performances of all-*d*-metal Ni-Co-Mn-Ti alloys through developing RF regression model. Such model achieves high accuracy in predicting structural parameters (e.g., cell volume, tetragonality), while showing moderate accuracy in estimating total magnetic moments across austenitic and martensitic phases, primarily due to the inherent difficulty in accurately resolving antiferromagnetic ground states and their complex magnetic interactions. By prioritizing phase transition, an explainable XGBT model to predict MCE of NiMn-based alloys was developed by Tang,¹¹⁹ providing a systematic mapping of MCE performance on Ni-Mn-In-Cu and Ni-Mn-In-Co alloys. The results demonstrated that an enhancement of $\sim 12\%$ for XGBT model was actualized when taking phase transition into account.

In addition to these typical magnetocaloric materials, research workers also explored and developed various magnetic refrigeration systems based on ML techniques, including REB₂-type alloys,¹²⁰ RE₂X₂Y ternary compounds,¹²¹ MgCu₂-type compounds,¹²² and

other materials systems,¹²³ which greatly promote the development of green refrigeration technology. Typically, Murgatroyd *et al.*¹²⁴ identified the change of *c/a* ratio between parent and child MnZnSb compounds via an improved RF model with input features including the compositions of both structures and the lattice parameters of the parent compound. The ML is not only used to predict magnetocaloric properties, but also employed to guide the distinguishment of phase transition types and phase structure,¹²⁵ the prediction of large magnetic moment materials,⁸⁹ as well as the design of magnetic refrigeration devices.¹²⁶

It should be mentioned that some other caloric effects also play significant roles in solid-state cooling technologies, such as elastocaloric and barocaloric effects. These technologies leverage the thermal response of materials to external stimuli (mechanical stress, pressure, and electric field) to achieve cooling without the use of harmful refrigerants. The elastocaloric effect is driven by latent heat exchange during stress-induced phase transitions, while the barocaloric effect exploits pressure-driven entropy changes in materials. Previous studies demonstrate that a large entropy change can be obtained under a relatively moderate pressure in Ni-Mn-In and Mn-Ga-N.^{127,128} Currently, the adopted cooling substances based on elastocaloric and barocaloric refrigerations are mainly NiTi-based superelastic shape-memory alloys.¹²⁹ Undoubtedly, many elastocaloric and barocaloric compounds with large entropy change are exploited progressively by the guidance of ML, such as NiTi-based alloys,¹³⁰ NiMn-based alloys,¹³¹ MnCu,¹³² and Cu-Al-based compounds.¹³³ It is reported that Kang and Elbel¹³⁴ proposed a novel packed rod bed regenerator for caloric cycles, which were optimized by using an artificial neural network-genetic algorithm (ANN-GA) and additive manufacturing. By further validating based on ML with computational fluid dynamics (CFD) simulations, the caloric cycles showed improved heat transfer and reduced pressure drop. The optimized regenerator enhanced system efficiency by 4.7% and cooling capacity by 13% compared to traditional parallel plate designs, offering potential for energy-efficient magnetic refrigeration. By integrating ML with computational and experimental approaches, researchers can uncover new materials with enhanced caloric effects, paving the way for the commercialization of efficient and sustainable solid-state cooling systems. This synergy between ML and caloric materials holds great promise for revolutionizing the future of refrigeration and thermal management.

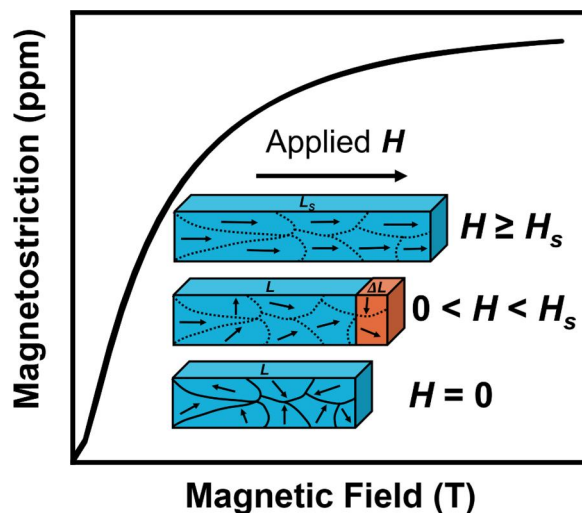


FIG. 8. Schematic for mechanism of magnetostrictive effect.

B. ML-aided design of magnetostrictive materials

Magnetostrictive effect refers to a phenomenon that the ferromagnetic materials undergo reversible deformation in response to an applied magnetic field. Such effect arises from the reorientation of magnetic domains which induces strain in magnetic materials,¹³⁵ as indicated in Fig. 8. At present, the most well-known magnetostrictive materials include Terfenol-D ($\text{Tb}_x\text{Dy}_{1-x}\text{Fe}_y$, $0.3 \leq x \leq 0.34$, $1.95 \leq y \leq 2$) compounds,^{136,137} and Galfenol (Fe-Ga) alloys.^{138,139} These materials are extensively used in sensors, actuators, and energy harvesting devices due to the ability to convert magnetic energy into mechanical energy and vice versa. Terfenol-D exhibits a giant magnetostriction up to 2000 ppm but is limited by the high cost and scarcity of rare-earth elements. In contrast, Galfenol offers moderate magnetostriction of 400 ppm with better mechanical properties and lower

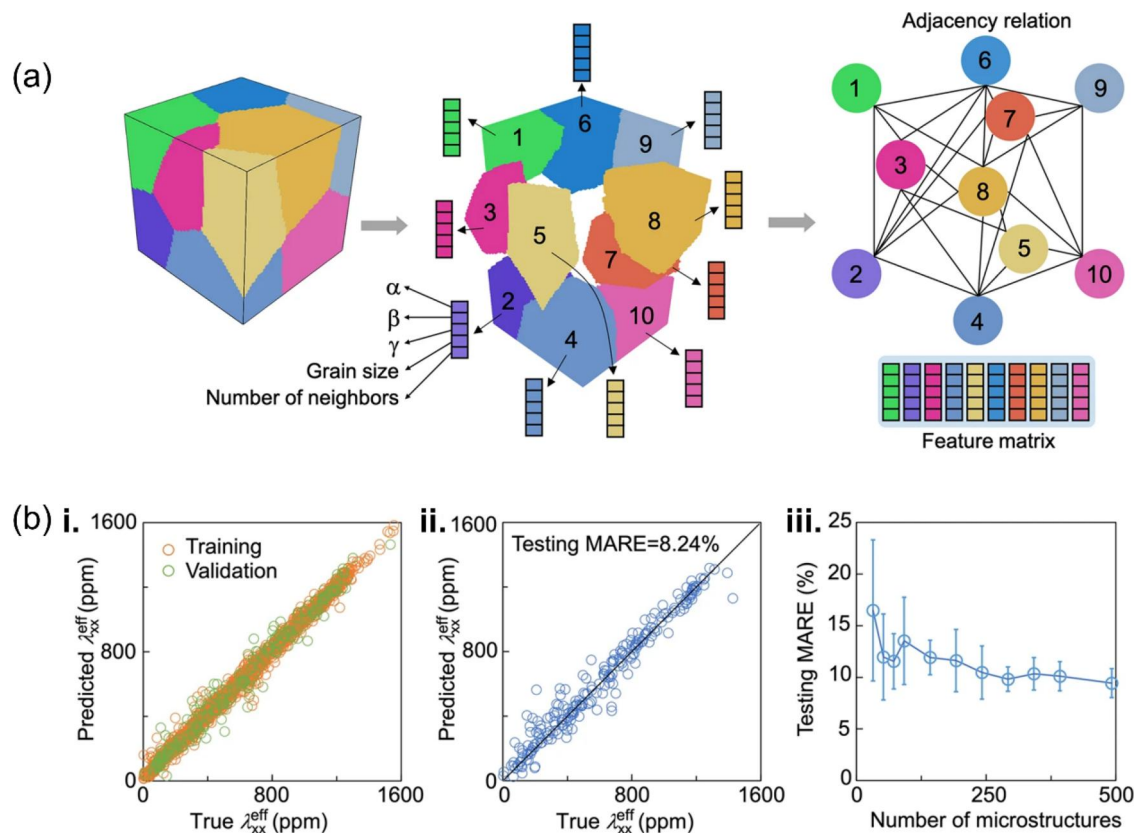


FIG. 9. (a) Representation of conversion from microstructure to graph input based on GNN model. (b) (i) Predictive performance of training and validation datasets for GNN model. (ii) Predictive performance of the independent testing dataset. (iii) Testing MARE of the GNN models trained based on datasets of different sizes. Reproduced with permission from Dai *et al.*, npj Comput. Mater. **7**, 103 (2021). Copyright 2021 Authors, licensed under a Creative Commons Attribution (CC BY) license.¹⁶

cost, making it suitable for applications requiring durability and precision.

For the last decades, many efforts have been devoted to addressing the challenges related to hysteresis, brittleness, and temperature sensitivity, such as the elements doping,¹⁴⁰ the process optimization,¹⁴¹ and the computer-assisted design based on PF simulation or DFT calculation.¹⁴² However, the current mainstream method of exploiting high-performance magnetostrictive materials is still trial-and-error approach in experiment through the scientific intuition of researchers, which inevitably leads to the soaring of costs.

ML models have been used to explore alternative compositions with reduced rare-earth content while maintaining high magnetostrictive performance. Dai *et al.*¹⁶ constructed a GNN model for predicting the properties of polycrystalline materials, focusing on the magnetostriction of Terfenol-D alloys. By representing microstructures as graphs with grain features and adjacency relations, the model achieves low prediction error ($\sim 10\%$) and enables interpretable, grain-level feature importance analysis, outperforming traditional CNN-based approaches in computational efficiency. Figure 9(a) manifests representation of conversion from microstructure to graph input. Five features constitute vector of each grain: three Euler angles (α , β , γ) for orientation, grain size in voxels, and the number of neighboring grains. A

separate adjacency matrix captures neighboring relations between grains, connected solid lines visually in right plot of Fig. 9(a). Figure 9(b) illustrates the property prediction and the macro average relative error (MARE) of the GNN models trained using datasets of different sizes. Furthermore, they also introduced the integrated gradient method to interpret the model predictions, revealing that the grain orientation plays a more significant role than grain size in determining magnetostriction.

Bayesian optimization efficiently finds optimal hyperparameters using probabilistic models, while principal component analysis (PCA) reduces data dimensionality by identifying principal components that capture maximum variance. Based on Bayesian optimization, Will-Cole *et al.*¹⁴³ employed optimized FeGaB and FeGaC thin films, targeting magnetostriction and ferromagnetic resonance (FMR) linewidth, which reduced required experiments by 50% compared to traditional methods. In addition, Lira Foggiatto *et al.*¹⁴⁴ investigated magnetostriction mechanisms in Fe–Ga alloy single crystals using fast Fourier transform (FFT) and unsupervised machine learning. By PCA decomposition of the MOKE images, the study distinguished 180° and 90° domain wall motions, linking them to magnetostriction variations. The results revealed complex correlations between magnetic domains and material properties, aiding future magnetostrictive material design.

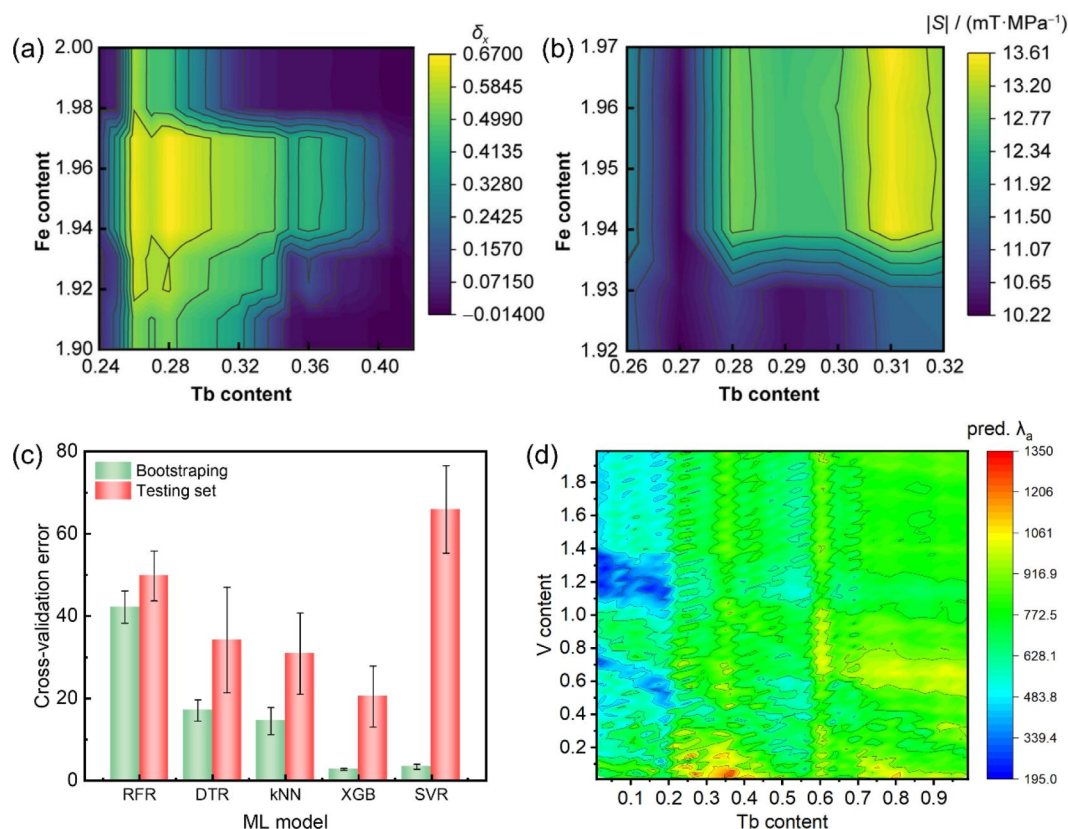


FIG. 10. (a) Colormap of low-field response (δ_x) and (b) piezomagnetic sensitivity corresponding to the content of Tb and Fe in $\text{Tb}_x\text{Dy}_{1-x}\text{Fe}_y$. Reproduced with permission from Gong *et al.*, *Rare Met.* **43**, 2251–2262 (2024). Copyright 2024 Springer Nature.²⁰ (c) Cross-validation error of models obtained by bootstrapping resampling method through 100 iterations. (d) Predicted magnetostriction heatmap of Tb–Dy–Fe–V alloys. Reproduced with permission from Hu *et al.*, *Mater. Des.* **252**, 113799 (2025). Copyright 2025 Authors, licensed under a Creative Commons Attribution (CC BY) license.¹⁹

In the bargain, an improved backpropagation neural network (BPNN) model combining Levenberg–Marquardt algorithm and particle swarm optimization was proposed by Zhou *et al.*,¹⁴⁵ predicting anisotropic magnetostriction in electrical steel sheets. Similarly, Wang *et al.*¹⁴⁶ also proposed an optimized BPNN model to predict anisotropic magnetostriction in grain-oriented electrical steel sheets (ESS) under DC bias. Through using magnetostrictive electromagnetic acoustic transducer (EMAT) signals and BPNN, a nondestructive method was constructed to measure mechanical properties of ferro-magnetic materials.^{147,148} Characteristic parameters from EMAT curves correlate strongly with material properties, achieving over 94% prediction accuracy. Lately, Gong *et al.*²⁰ adopted the GBRT and high-throughput micromagnetic simulation approach to optimize the magnetoelastic properties of $\text{Tb}_x\text{Dy}_{1-x}\text{Fe}_y$ giant magnetostrictive materials. According to micromagnetic simulation results, the low-field response (δ_x) exceeding 0.48 is located in the compositional region of $0.26 \leq x \leq 0.32$ and $1.92 \leq y \leq 1.97$ exhibited in Fig. 10(a), which presumably stems from magnetocrystalline anisotropy compensation with large magnetostriction near morphotropic phase boundary. In Fig. 10(b), it is discovered that the maximum piezomagnetic sensitivity ($S = \frac{d\mu_0 M}{d\sigma}$) can reach $10.22\text{--}13.61 \text{ mT} \cdot \text{MPa}^{-1}$ in the optimized composition range of $\delta_x \geq 0.48$. Afterward, they presented

a CNN model for high-precision stress/strain detection using magnetoelastic coupling in $\text{Tb}_{0.27}\text{Dy}_{0.73}\text{Fe}_2$, further achieving micro-strain resolution ($\sim 10^{-5}$) with low hysteresis to enable real-time monitoring.¹⁴⁹ The network extracts magnetic domain features, offering a novel approach for nano/micro-scale stress sensing in applications like geological monitoring and aerospace.

In fact, it can be found that the research on predicting magnetostrictive performance is rather rare. One of the most important reasons may be the difficulty in building a database because the MFAs exhibit intricate magnetic and crystal structures.^{150–152} To solve this dilemma, we created a dataset by constructing composition–physical parameters–magnetostriction relationship. Based on the home-built dataset, a physics-informed interpretable ML strategy is used to design Laves-phase REFe_2 -type alloys with superior magnetostrictive performance.¹⁹ The cross-validation errors of different ML models are determined by bootstrapping resampling method through 100 iterations. The predicted errors of these models by 10-fold cross validation are plotted in Fig. 10(c), which demonstrate that the XGBoost model exhibits the smaller error compared to other models. By the XGBoost model, the optimal composition ranges for Tb–Dy–Fe–V alloys are determined by the predicted heatmap of magnetostriction distribution, as manifested in Fig. 10(d). Additionally, the SHAP

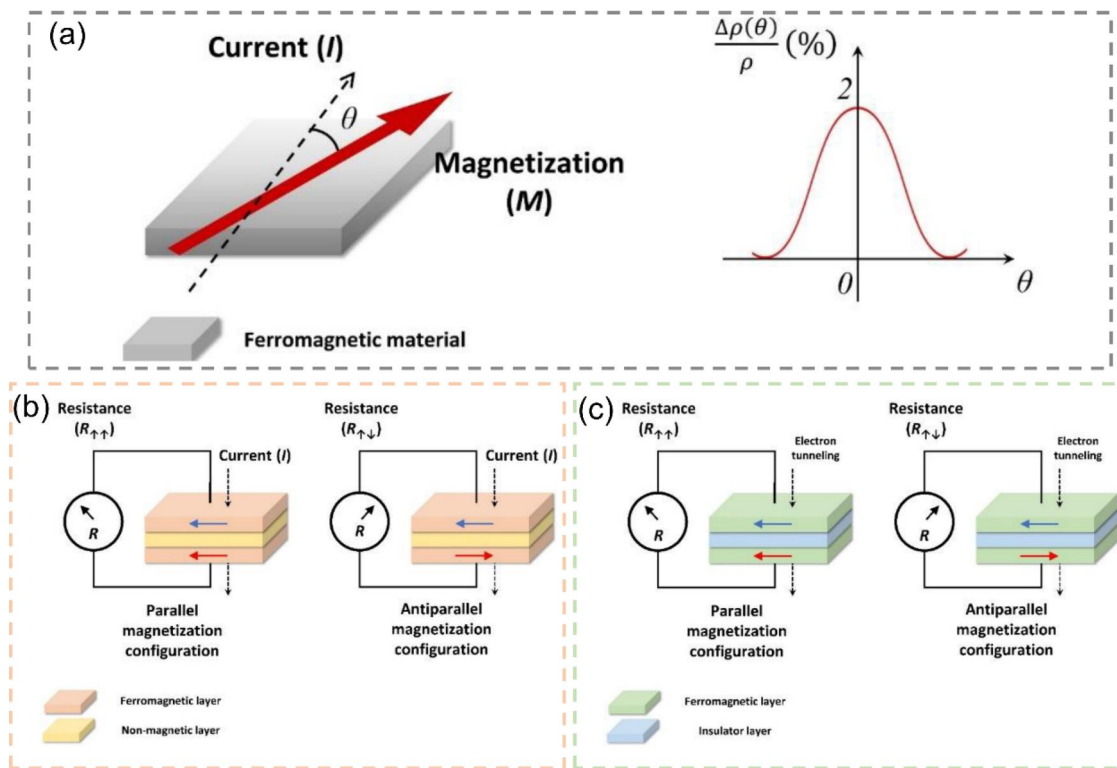


FIG. 11. (a) Schematics for (a) anisotropic magnetoresistance, (b) giant magnetoresistance, and (c) tunneling magnetoresistance. Reproduced with permission from Yang *et al.*, *Chemosensors* **9**, 211 (2021). Copyright 2021 Authors, licensed under a Creative Commons Attribution (CC BY) license.¹⁵⁷

indicated that the volume magnetic susceptibility and the bulk modulus have significant influences on magnetostriction. Recent findings further explored the relationship between composition and property of REFe₂-type compounds by constructing a DL-based predictor, which exhibited superior predictive ability compared to traditional models.¹⁵³ These approaches accelerate the discovery of high-performance magnetostrictive materials, offering a data-driven pathway for functional material design.

C. ML-aided design of magnetoresistance materials

Magnetoresistance refers to the change in electrical resistance of a magnetic material under an applied magnetic field, which is related to the mechanism of electron scattering.¹⁵⁴ In general, the magnetic and crystal structures of materials are changed when subjected to an external magnetic field, whereas the transport capability of electrons strongly depends on the arrangement and spin configuration of atoms, which induce the variation of electrical resistivity.¹⁵⁵ Based on this effect, the magnetoresistance is commonly categorized into the anisotropic magnetoresistance (AMR), the giant magnetoresistance (GMR), the tunneling magnetoresistance (TMR), and the colossal magnetoresistance (CMR). The AMR is the earliest discovered effect, where resistance varies with the angle between the current and magnetization, commonly observed in ferromagnetic metals like nickel and iron.¹⁵⁶ Figures 11(a)–11(c) exhibit the basic principle of AMR, GMR, and TMR.¹⁵⁷ The GMR, discovered in the late 1980s, occurs in

multilayered structures of alternating ferromagnetic and nonmagnetic layers,¹⁵⁸ leading to significant resistance changes due to spin-dependent electron scattering.¹⁵⁷ The GMR revolutionizes data storage, enabling high-density hard drives. Equally, the TMR involves electron tunneling through an insulating barrier between ferromagnetic layers, with resistance depending on the relative magnetization alignment,^{157,159} which is key to magnetic tunnel junctions in magnetoresistance random access memory (MRAM). The CMR was observed in perovskite manganites,^{160,161} where the resistance changes dramatically near magnetic phase transitions driven by complex electron-lattice interactions. These materials continue to drive advancements in spintronics and memory technologies.

In the past few years, the advent of ML has revolutionized the area by enabling rapid prediction and design of magnetoresistance materials with tailored properties.¹⁶² Ju *et al.*¹⁶³ employed Bayesian optimization and LASSO regression combined with DFT calculations to investigate the TMR effect in Fe/disordered-MgAl₂O₄/Fe magnetic tunnel junctions (MTJs). Figure 12(a) presents the averaged convergence trends, clearly demonstrating the superior efficiency of Bayesian optimization compared to random search. As the LASSO regularization strength α increases, the in-plane Al–Al distance (d_{XY} Al–Al) remains the most significant descriptor for characterizing TMR shown in Fig. 12(b). It can be also found in Fig. 12(c) that LASSO predictions with $\alpha = 0.1$ qualitatively match first-principles TMR ratios, since descriptors accounting for interfacial structures are

incorporated alongside cation-site atomic distances. Furthermore, Zhang *et al.*¹⁶⁴ developed deep-learning neural networks to enable large-scale simulations of electronic phase separation in the single-band double-exchange model, which undertakes a significant role in CMR materials. As exhibited in Fig. 12(d), a six-layer NN model with excellent performance was established to predict components of local exchange forces H_i . The NN model was then integrated into Landau-Lifshitz-Gilbert (LLG) dynamics to perform a quench simulation on a 30×30 lattice from random spins down to $T = 0.022$, where parallel simulations using exact diagonalization (ED) produce nearly identical spin-spin correlation functions across different electron fillings, as depicted in right plot of Fig. 12(d), confirming the accuracy and transferability of model in dynamical simulations. The study revealed a freezing behavior where the doped holes segregated into ferromagnetic clusters, creating confining potentials that reduce hole mobility and arrest phase separation.

Kurniawan *et al.*¹⁶⁵ explored highly spin-polarized Heusler alloys at finite temperatures using machine learning and first-principles calculations, which identified promising materials like $\text{Co}_2\text{MnGa}_{0.2}\text{As}_{0.8}$ and $\text{Co}_2\text{FeAl}_{0.4}\text{Sn}_{0.6}$ retaining high spin polarization at room temperature. This work also highlighted the importance of Fermi level tuning and alloy mixing to improve temperature-dependent spin polarization. Furthermore, the effect of Fermi surface matching at Co-Fe/Cu interfaces on GMR was investigated by utilizing the combinatorial techniques of ML and first-principles calculations, eventually observing

the highest GMR ratio of 17.5% in $\text{Co}_{0.54}\text{Fe}_{0.46}$.¹⁶⁶ ML force-field model is an important way for simulating the adiabatic dynamics of cooperative Jahn-Teller (JT) systems.^{167,168} As shown in Fig. 12(e), the ML model enabled large-scale Langevin dynamics simulations, which revealed freezing behaviors in the late-stage coarsening of orbital domains, highlighting its potential for multiscale dynamical modeling of correlated electron systems.¹⁶⁷

ML has become a strong tool for the design and optimization of magnetoresistance materials, offering a promising pathway for the development of next-generation spintronic devices. Wang *et al.*¹⁶⁹ proposed a portable magnetic stripe authentication system using a TMR sensor for banknote verification under the assistance of different denomination classification algorithms including peak spacing, BPNN, and SVM. The system achieves high accuracy and reliability, offering a practical solution for handheld devices in detecting counterfeit currency. Xin *et al.*¹⁷⁰ investigated the influence of various factors on a NN-based current transformer algorithm using a GMR sensor array. They analyzed the impact of sensor count, noise, wire eccentricity, and inclination on accuracy. The results demonstrate the algorithm's robustness and provide design guidelines for optimizing sensor arrays in practical applications. Shukla *et al.*¹⁷¹ explored the statistical variation analysis of voltage-gated spin-orbit torque magnetic tunnel junctions (VGSOT-MTJs) using ML. They proposed an ML-assisted framework to predict device performance, comparing KNN, RF, and NN regressors. The RF model outperforms others in

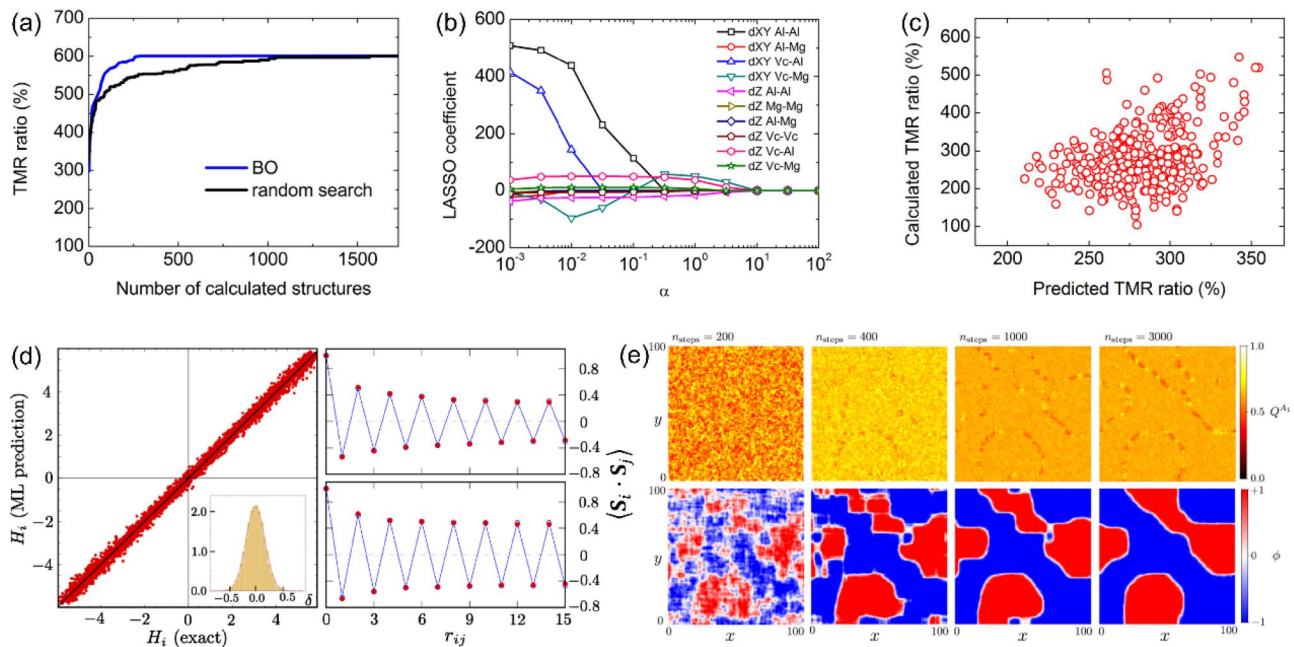


FIG. 12. (a) Comparison of Bayesian optimization with random search. (b) LASSO regression coefficients characterizing the TMR ratio related to interatomic distances. (c) Comparison of TMR ratio obtained from DFT calculations and predicted by LASSO with $\alpha = 0.1$. Reproduced with permission from Ju *et al.*, Phys. Rev. Res. **2**, 023187 (2020). Copyright 2020 Authors, licensed under a Creative Commons Attribution (CC BY) license.¹⁶³ (d) Components of local exchange forces H_i predicted by trained NN model vs the exact results on test datasets. Right: spin-spin correlation functions $\langle S_i \cdot S_j \rangle$ vs $r_{ij} = |r_j - r_i|$ along the x direction at electron filling fraction $n = 0.485$ and $n = 0.475$. Reproduced with permission from Zhang *et al.*, Phys. Rev. Lett. **127**, 146401 (2021). Copyright 2021 American Physical Society.¹⁶⁴ (e) Initial random state quenched to $T = 0.001$ at $t = 0$ ($n_{\text{step}} = 0$). ML-Langevin simulations track relaxation. Orbital domains ($\phi = \pm 1$) shown in red/blue ($\Delta t = 0.05\tau_0$). Q^{A_1} breathing modes and ϕ_i evolution visualized. Reproduced with permission from Ghosh *et al.*, Phys. Rev. Mater. **8**, 123602 (2024). Copyright 2024 American Physical Society.¹⁶⁷

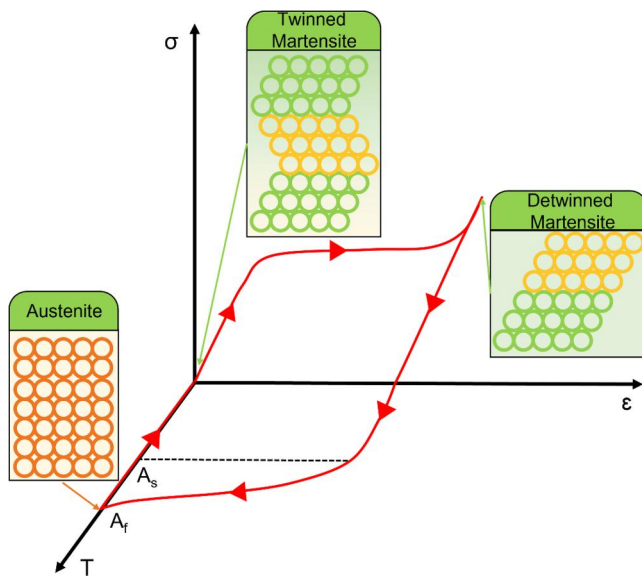


FIG. 13. Schematic for mechanism of shape memory effect.

accuracy and efficiency, offering a faster alternative to traditional SPICE simulations for optimizing VGSOT-MTJ designs.

Additionally, Sehgal *et al.*¹⁷² adopted an ANN-based surrogate model to analyze variability in multilevel SOT-MRAM, predicting the impact of device parameter variations on resistance and write energy. Dai *et al.*¹⁷³ also employed a NN-based algorithm for current transformers using GMR sensor arrays to measure magnetic fields and infer current magnitudes. Moreover, Xu *et al.*¹⁷⁴ developed a novel eddy current testing method using a spin Hall magnetoresistance (SMR) sensor, which operates under AC bias, eliminating the need for demodulation. The SMR sensor, combined with PCA and a decision tree model, effectively detects and classifies metal cracks with high accuracy, thereby simplifying the eddy current testing system. According to these applications, it can be found that magnetoresistance is revolutionizing application field of spintronic devices ranging from magnetic field detection to advanced memory and computing technologies due to their high sensitivity, low power consumption, and the ability to operate at ultrafast speeds.

D. ML-aided design of shape memory alloys

Shape memory alloys (SMAs) are a class of smart materials that exhibit the unique ability to recover their original shape after deformation when subjected to a specific thermal, magnetic, or mechanical stimulus. Such novel property is derived from a reversible martensitic phase transformation, i.e., shape memory effect shown in Fig. 13, which is discovered in Cd–Au alloys.¹⁷⁵ Owing to this property, SMAs are regarded as alternative materials in diverse areas like automotive, robotic actuators, connectors, and smart wings.

In recent a review, Kim *et al.*¹⁷⁶ compared the characteristics of thermal SMAs (TSMAs) and magnetic SMAs (MSMAs) based on the actuation performance indices. For TSMAs, a deformation cycle starts with the conversion of twinned martensite to detwinned martensite under external stress. Upon heating above the austenite start

temperature (A_s), the martensite phase is fully converted into the austenite type at the austenite finish temperature (A_f).¹⁷⁷ Conversely, the TSMAs undergoes a transformation from austenite to martensite upon cooling to the martensite start temperature (M_s), entirely completing the transformation at the martensite finish temperature (M_f). Due to the broad transformation temperature range, reliable actuation and affordability, TSMAs have been commercialized, typically in wire and plate forms. However, TSMAs depend on heat for actuation, resulting in low energy efficiency, limited bandwidth, and slower actuation speeds at larger scales.¹⁷⁶ Similarly, under an applied magnetic field, MSMAs can also be induced a reverse transformation between martensite and austenite. Since the magnetic shape memory effect (MSME) was reported in Fe–Ni alloys, researchers have paid more attention to enhance the recoverable strain and expand the operating temperature range.¹⁷⁸ Unlike TSMAs, MSMAs are still in the development stage, and the commercialized application is immature. Presently, the design and optimization of MSMAs with desired transformation temperatures and mechanical properties remain challenging due to the complex interplay among composition, microstructure, and processing conditions.

Recent advances in ML have opened new avenues for accelerating the discovery and optimization of SMAs.¹⁷⁹ In particular, NiTi-based alloys hold immense promise as next-generation multifunctional materials, integrating a diverse suite of vital functional and mechanical properties. A novel framework of multi-objective Bayesian optimization was proposed to discover precipitation-strengthened NiTi SMAs.¹⁸⁰ Within the framework, a GPR model acted as the probabilistic surrogate during the machine learning step, which accurately predicted M_s (3.2 °C for prediction and 3.1 °C experiment). Equally, Li *et al.*¹⁸¹ presented a machine learning and multi-objective optimization approach to enhance properties of NiTi shape memory alloy via laser powder bed fusion (LPBF), achieving high strength of 783 MPa, elongation of 13.7%, and low thermal hysteresis of 15.1 K. The triangular heatmaps of predictive hysteresis and M_s temperature, subsequently, were created in a crack-free NiTiCu alloy via CALPHAD and ML methods.²³ Figure 14(a) indicates a general reduction in thermal hysteresis across the 3–11 at. % Cu range, exhibiting a pronounced local minimum near 9 at. % Cu (excluding high-uncertainty corners). Correspondingly, Fig. 14(b) presents the predicted M_s temperature for the B2-B19 transition, which varies between 261.8 and 349.4 K over the composition space. To predict transformation temperatures and hysteresis of NiTiHf-SMAs, Liu *et al.*¹⁸ developed a physics-informed GPR model by incorporating both elemental features and engineered heat treatment descriptors based on phase transformation kinetics. The model achieved R^2 values of 0.92 for mean transformation temperature and 0.68 for hysteresis with low prediction uncertainty, significantly improving design accuracy over composition-only approaches. It also noticed that performance degraded at edge cases like Ni-lean compositions due to data sparsity, highlighting the need for broader training data coverage beyond Ni-rich alloys.

Machaka and Radingoana¹⁸² demonstrated the successful prediction of narrow thermal hysteresis in TiNi-based high-temperature SMAs using an XGBoost model on a small experimental dataset. The key advantage is the integration of explainable AI (i.e., relative feature importance, accumulated local effects and SHAP) to interpret predictions, revealing that Ni, Cu, and Zr content are the most influential

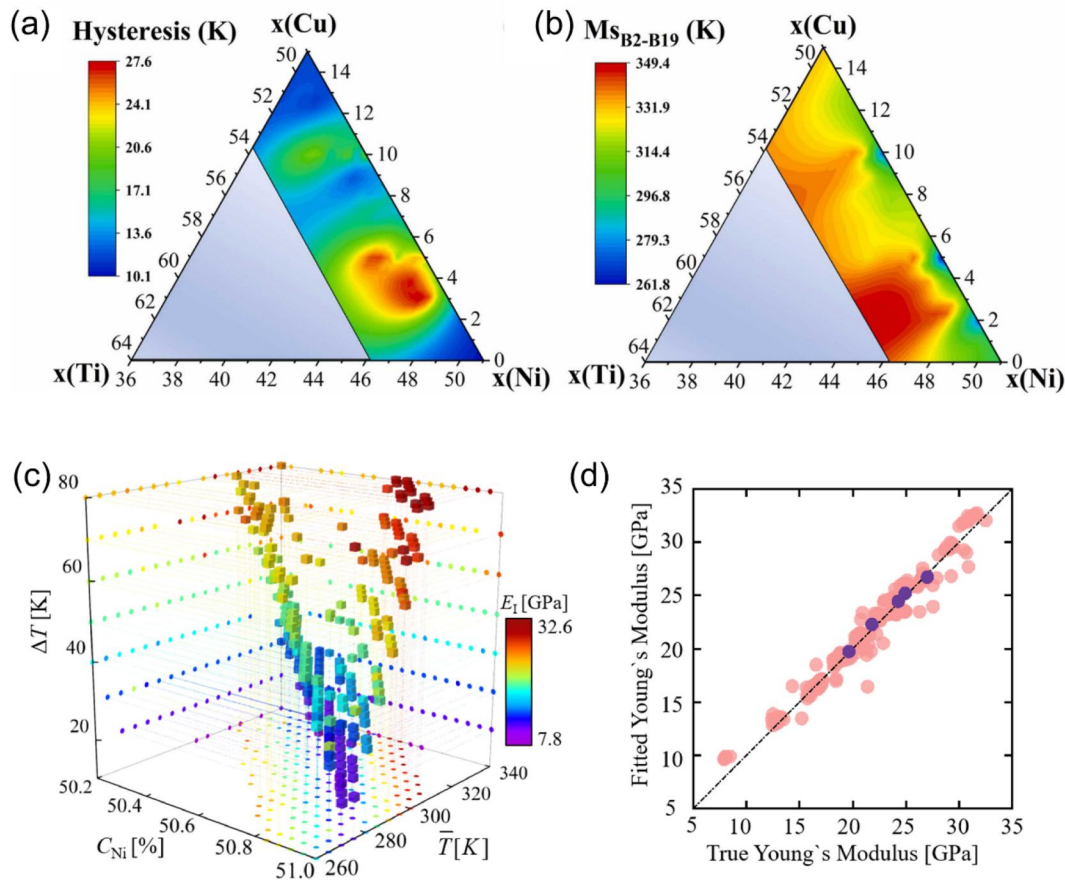


FIG. 14. Triangle heatmaps of (a) predicted thermal hysteresis, and (b) predicted M_s (B2-B19) across the composition range. Reproduced with permission from Li *et al.*, *Addit. Manuf.* **92**, 104386 (2024). Copyright 2024 Elsevier.²³ (c) Determination of specimen Young's modulus via high-throughput phase-field simulation. (d) A comparison of Young's modulus from phase-field simulations vs machine learning predictions. Reproduced with permission from Xu *et al.*, *Int. J. Mech. Sci.* **259**, 108592 (2023). Copyright 2023 Elsevier.¹⁸⁵

features. Correspondingly, they provided two curated datasets: a main set of 190 experimental TiNi-based high-temperature SMAs for narrow thermal hysteresis, and an auxiliary set from an explainable XGBoost model. This provides an interpretable, physics-aligned data resource for accelerating alloy design and validating new candidates.¹⁸³ Moreover, Honrao *et al.*¹⁸⁴ constructed machine learning models using an 8000-alloy NiTi dataset to predict transformation temperatures (MAE $\approx 14.8^\circ\text{C}$), hysteresis (7.2°C), and transformation strain (0.36%). A major strength is the capability of model to predict key shape memory properties and elucidate processing effects for a vast range of multi-component alloys. By combining high-throughput PF simulations and ANN algorithm, Xu *et al.*²⁵ designed TiNb nanocomposites with ultralow modulus of ~ 16 GPa, linear superelasticity, and near-zero hysteresis. The optimized Nb-rich matrix/Nb-lean nanofiller structure enables continuous martensitic transformation. Thereafter, they integrated active learning with PF simulations to demonstrate that temperature gradients in NiTi alloys enable linear superelasticity and ultralow modulus by modulating martensitic transformation, while the SISSO (sure independence screening and sparsifying operator) derived models established

explicit Young's modulus expressions for practical temperature engineering of SMAs.¹⁸⁵ As illustrated in Fig. 14(c), despite an initial reduction under a temperature gradient compared to a uniform temperature, Young's modulus exhibited an overall increasing trend with temperature difference. From the training data, the SISSO algorithm screened an optimal descriptor linking the mechanical properties to the temperature field, which obtained a relationship described as $E_I = 5.9464 + 0.0306 \times (\Delta T \times (\bar{T} - \Delta T)) - 1.4866 \times ((\bar{T} - \Delta T) \times (\Delta T / C_{Ni}))$, where E_I is Young's modulus, C_{Ni} is concentration of Ni, ΔT and $\bar{T}$ represents temperature difference and average temperature, respectively. Figure 14(d) manifests the performance of ML model with an R^2 value of 0.95. Specifically, the alignment of the five newly sampled specimens (blue circles) along the diagonal line confirms the model's reliability in predicting mechanical properties. Meanwhile, Tian *et al.*²⁴ screened NiTi-based SMAs with narrow thermal hysteresis combining XGBoost and DFT calculations, identifying potential compositions with minimal hysteresis for elastocaloric applications. In addition, they also predicted the martensitic transformation temperature of NiMnSn-based

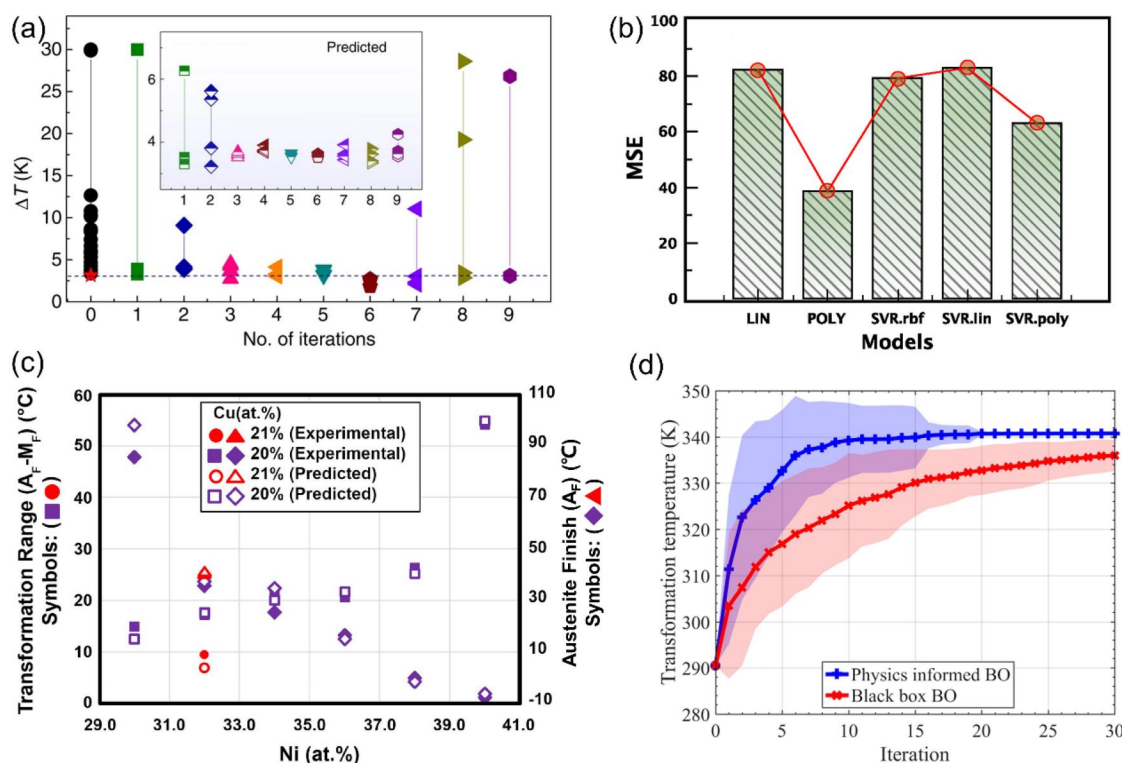


FIG. 15. (a) Experimental measurements for thermal hysteresis vs number of iterations based on adaptive design loop compared with the predictions of inset. Reproduced with permission from Xue *et al.*, Nat. Commun. 7, 11241 (2016). Copyright 2016 Authors, licensed under a Creative Commons Attribution (CC BY) license.¹⁹³ (b) Comparison of different models in terms of the MSE on the test dataset. Reproduced with permission from Xue *et al.*, Acta Mater. 125, 532–541 (2017). Copyright 2017 Elsevier.²¹ (c) Comparison between predicted and DSC-measured martensitic transformation ranges and austenite finish temperatures for NiTiCu alloy. Reproduced with permission from Trehern *et al.*, Acta Mater. 228, 117751 (2022). Copyright 2022 Elsevier.²² (d) Maximum transformation temperature determined based on physics-informed and black-box BO. Reproduced with permission from Khatamsaz *et al.*, npj Comput. Mater. 9, 221 (2023). Copyright 2023 Authors, licensed under a Creative Commons Attribution (CC BY) license.¹⁹⁴

MSMAs,¹⁸⁶ the thermal hysteresis of high-entropy SMAs,¹⁸⁷ and the elastocaloric performance of NiTi-based alloys.¹⁸⁸

The phase transition temperature (T_p) is a fundamental property of magnetic materials, dictating the magnetic ordering and transition behavior.¹⁸⁹ The manipulation of transition temperature T_p is a crucial way for tailoring the properties of SMAs. Xue *et al.*^{15,190–192} established an adaptive design strategy to efficiently determine the transformation temperatures and reduce the thermal hysteresis in Ti–Pd–Cr and NiTi-based SMAs. Figure 15(a) demonstrates the results by performing nine design iterations, which illustrates the evolution of the experimental (and predicted, inset) thermal hysteresis (ΔT) across successive iterations.¹⁹³ They further predicted the T_p of NiTi-based SMAs using a three-feature polynomial regression model, validated with an adaptive design loop to efficiently guide synthesis toward targeted properties.²¹ It is observed in Fig. 15(b) that an evaluation of MSE across different models on the test dataset demonstrated that the polynomial model achieved the best performance.

In recent years, research on predicting NiTi-based alloys using various ML approaches has shown explosive growth. Trehern *et al.*²² successfully applied the artificial intelligence materials selection (AIMS) framework to discover NiTi-based SMAs with superior functional properties. A comprehensive dataset (including composition,

processing, and stress) with machine learning was integrated to predict the alloy $\text{Ni}_{32}\text{Ti}_{47}\text{Cu}_{21}$, which exhibited a minimal transformation range ($A_f - M_s$) of approximately 9.5 °C and excellent cyclic stability under applied stress. In addition, the designed $\text{Ni}_{32}\text{Ti}_{47}\text{Cu}_{21}$ demonstrates a distinct advantage with a transformation range that is 5.4 °C narrower than the next best-performing Ti-lean NiTiCu sample, as described in Fig. 15(c). Afterward, Khatamsaz *et al.*¹⁹⁴ proposed a physics-informed Bayesian optimization (PIBO) approach by integrating physical knowledge into Gaussian process kernels. In the design of NiTi SMAs, PIBO possessed superior efficiency which reached the target transformation temperature of ~ 340 K in just 15 iterations, while the black-box BO persistently fell below this performance benchmark throughout all 30 iterations, which is exhibited in Fig. 15(d). This study is beneficial for enhancing data efficiency, reducing modeling uncertainty, and enabling broader exploration of the optimal processing space. For NiTi-based SMAs, a number of researches were performed to regulate the transformation temperatures and the physical properties by the assistance of different ML models.^{75,195–205}

As an outstanding refrigerant, SMAs has been used in the design of elastocaloric refrigeration devices.^{12,111} Accordingly, ML technology is adopted to accelerate the discovery of SMAs with high elastocaloric entropy change and temperature change.^{133,188,206} Song *et al.*²⁰⁷

prognosticated the electrochemical machining outcomes for SMAs based on DNN, demonstrating higher accuracy compared to traditional methods like response surface method, Taguchi and regression analysis. In addition, Zadeh *et al.*²⁰⁸ developed an interpretable piecewise linear regression model to distinguish the martensitic transformation type in NiTi compounds. The lattice constant can significantly influence the phase transformation behavior and the mechanical properties of SMAs, as it directly affects the stability of the austenite and martensite phases. Some properties like the lattice constants were predicted accordingly based on various ML models for NiTi-based high-entropy SMAs.²⁰⁹ In practical applications, SMAs offer lightweight, compact, and efficient actuation with high energy density and reversible deformation, ideal for precise and durable device applications. For example, Dutta *et al.*²¹⁰ developed an automated AI-driven system for characterizing NiTi-based SMAs foils produced via planar flow casting. Using computer vision and XGBoost regression, the system predicted the actuation angles (95% accuracy) and generated forces (99.5% accuracy) from infrared video data. Assisted by ML, recent studies are devoted to actualizing the optimization and tuning equipment parameters and promoting the rapid development of SMA-actuated smart devices.^{211–214} These ML-driven approaches not only enhance the predictive capability for SMAs properties but also provide insights into the underlying physical mechanisms governing their behavior.

IV. CHALLENGES AND OPPORTUNITIES

The convergence of ML techniques and big data have ushered in a new paradigm for accelerating the discovery and optimization of MFAs. Nevertheless, the data-driven approach is still at a nascent stage, thus facing many challenges in unearthing novel MFAs with tailored functionalities. Accordingly, some challenges and opportunities for future research directions are elaborated in this section.

A. Dataset development

The prediction accuracy of the ML model is directly associated with the quality and quantity of dataset. For the area of MFAs, experimental data are typically small and fragmented. It can be found that the dataset adopted in ML-aided MFAs design basically derives from the published papers. This could produce profound influence on prediction results, since publications commonly report the materials with the best properties, whereas the failed results are neglected. Actually, the failed data also possess great value for training ML models, as demonstrated by using the reaction data of failed hydrothermal syntheses to predict reaction outcomes in inorganic-organic hybrid materials.²¹⁵ Consequently, it is essential to establish a data exchanging mechanism that achieves data sharing in AI and material research under the development of blockchain.

B. ML model interpretability

ML models operate as “black boxes” due to the difficulty in establishing relationship between input and output, especially in magnetic systems, leading to the un-interpretability of ML models. To overcome this issue, the feature importance is adopted to analyze the contribution of features to the model output based on the different methods such as Gini importance²¹⁶ and Permutation importance,²¹⁷ etc. However, the feature importance only qualitatively evaluates the impact of feature on prediction results. Hence, some promising

solutions should be employed including developing physics-informed neural networks that incorporate domain knowledge, applying explainable AI techniques like SHAP values and attention mechanisms, using surrogate interpretable models, and implementing hybrid approaches that combine ML with first-principles calculations. Recent work has shown success in applying these methods to predict properties in MFAs while maintaining interpretability. For example, Hu *et al.*¹⁹ provided a quantitative description for feature contribution by combining SHAP and Locally Weighted Scatterplot Smoothing curve. Jiang *et al.*²¹⁸ uncovered key physical quantities affecting prediction target based on feature selection and symbolic. These approaches improve significantly prediction accuracy of ML models.

C. Multiscale complexity of MFAs

As a category of functional materials, improving performance is the top priority for MFAs. However, MFAs exhibit complex behaviors spanning multiple length and time scales from atomic spin interactions (microscale) to domain wall dynamics (mesoscale) and bulk magnetic properties (macroscale). Such multiscale nature poses significant challenges for constructing relationships between feature factors and target attributes. As an example, there is no specific expression to correspond to the target attributes in magnetostriction due to the complex magnetic and crystal structures, thus difficulty in forming a mapping relationship between physical parameters and magnetostriction. This generates a significant obstacle to establishing ML datasets. It can be also found that research on ML-aided magnetostriction prediction is still insufficient. In the future, the ML-driven MFAs design lies in the development of hybrid approaches that combine physics-based models with data-driven techniques. For instance, incorporating domain knowledge into ML architecture, i.e., physics-informed neural networks, could enhance predictive accuracy and generalizability.

D. ML-aided inverse material design

In materials science, designing a new material requires producing or optimizing the target performance by determining the alloying composition in a vast search space, which belongs to a forward design process. In contrast, the inverse design reverses the research paradigm through starting with the desired target properties and identifying optimal compositions or structures that meet these targets.^{219–221} It is known that the forward mapping between material parameters and target performances can be clarified by the forms of experimental measurements or theoretical calculations, whereas the inverse design from target performances to material parameters cannot be acquired in the same way.²²² Surprisingly, ML is adept at tackling this kind of problem, formulating inverse design process via deep generative models in metal-organic framework materials,²²³ polycrystalline solids,²²⁴ and high-entropy bulk metallic glasses.²²⁵ The integration of advanced ML techniques with systematically expanded experimental datasets will facilitate autonomous image analysis, establishing robust correlations between compositional/microstructural characteristics and target MFA functionalities via data-driven predictive modeling.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Pengqiang Hu: Conceptualization (lead); Data curation (lead); Formal analysis (lead); Investigation (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (lead). **Chao Zhou:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Funding acquisition (equal); Project administration (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Sidan Ding:** Data curation (lead); Formal analysis (equal); Investigation (equal); Visualization (lead). **Ruisheng Zhang:** Data curation (equal); Formal analysis (lead); Investigation (lead); Visualization (equal). **Yizhe Ma:** Data curation (equal); Formal analysis (equal); Visualization (equal). **Kun Wang:** Conceptualization (equal); Formal analysis (equal). **Hongwen Chen:** Formal analysis (equal); Investigation (equal). **Zhiyong Dai:** Conceptualization (equal); Formal analysis (equal). **Rui Zhang:** Formal analysis (equal). **Xiaoqin Ke:** Formal analysis (supporting); Resources (supporting). **Zhengming Zhang:** Conceptualization (supporting); Formal analysis (equal). **Dunhui Wang:** Supervision (supporting). **Yao Liu:** Formal analysis (supporting); Resources (supporting). **Qiang Wang:** Resources (supporting); Supervision (equal). **Tie Liu:** Resources (supporting); Validation (equal). **Wenli Pei:** Formal analysis (supporting). **Tianli Zhang:** Formal analysis (equal); Funding acquisition (supporting). **Jingmin Wang:** Formal analysis (supporting). **Jie Zhu:** Formal analysis (equal). **Xuexu Gao:** Supervision (supporting). **Dan Huang:** Formal analysis (supporting); Funding acquisition (supporting). **Yin Zhang:** Formal analysis (supporting); Resources (supporting). **Fanghua Tian:** Formal analysis (supporting); Resources (supporting). **Kaiyan Cao:** Formal analysis (supporting). **Sen Yang:** Funding acquisition (equal); Project administration (equal); Resources (equal); Supervision (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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